INTEGRATED AUTOMATIC LICENSE PLATE RECOGNITION AND LIVE GPS SYSTEM AND SMART AUTHORIZATION

A PROJECT REPORT

Submitted by

GANESH.C (112721106006)

SARATHY.S (112721106021)

In partial fulfilment of the award of the degree

Of

BACHELOR OF ENGINEERING/TECHNOLOGY

IN

ELECTRONICS AND COMMUNICATION ENGINEERING

ST. PETER'S COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution)

ANNA UNIVERSISTY

CHENNAI-600025

MAY 2025

ST. PETER'S COLLEGE OF ENGINEERING AND TECHNOLOGY

(An Autonomous Institution)

BONAFIDE CERTIFICATE

Certified that the project report "**INTEGRATED AUTOMATIC LICENSE PLATE RECOGNITION AND LIVE GPS SYSTEM AND SMART AUTHORIZATION"** is a Bonafide work of **GANESH C (112721106006) and SARATHY S (112721106021),** who carried out the project work under my supervision.

SIGNATURE	**SIGNATURE**
Dr. A. SWARNALATHA, M.E., Ph.D.	**Mrs. R. ANUJA**
HEAD OF THE DEPARTMENT	**SUPERVISOR**
DEAN ACADEMICS	Associate Professor
Professor	Department of Electronics and Communication Engineering
Department of Electronics and Communication Engineering	St. Peter's College of Engineering and Technology,
St. Peter's College of Engineering and Technology,	Avadi, Chennai- 600054.
Avadi, Chennai- 600054.	

Submitted for viva-voce held on**……..at St. Peter's College of Engineering and Technology.**

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

We would like to begin by thanking the Almighty God for His abundant grace and blessings, which enabled us to complete this project successfully. We acknowledge the unwavering support and guidance provided throughout this journey with deep respect and gratitude.

We sincerely thank the Management of **St. Peter's College of Engineering and Technology** for offering us the platform and resources to carry out our project effectively. We express our heartfelt gratitude to our beloved Chairperson, Tmt. **Dr. Bhanumathi Thambidurai**, whose inspiring leadership and vision have always motivated us to strive for excellence.

We are truly grateful to our Principal**, Dr. S. Poornachandra, M.Tech., Ph.D.,** for his constant encouragement, support, and valuable guidance. We also sincerely thank **Dr. A. Swarnalatha, M.E., Ph.D.**, **Dean Academics,** Head of the Department of Electronics and Communication Engineering, for her motivation and insightful suggestions that greatly benefited our work.

We extend our profound appreciation to our internal guide, **Mrs. Anuja, Assistant Professor**, Department of Electronics and Communication Engineering, for her continuous support, expert advice, and thoughtful feedback throughout every phase of the project.

We also thank all our faculty members and technical staff for their timely help and cooperation, which contributed immensely to the successful completion of our project.

ABSTRACT

This project, titled **"Integrated Automatic License Plate Recognition and Live GPS System and Smart Authorization,"** is an innovative embedded system that seamlessly integrates three pivotal technologies: vehicle identification using Automatic License Plate Recognition (ALPR), real-time location tracking via GPS, and intelligent access control through QR code-based authorization.

The system employs a Raspberry Pi 4B coupled with a CCTV IP camera to capture high-quality images of vehicle License plates, which are processed using a custom-trained YOLOv8n model. This model effectively identifies and extracts characters from the images, enabling accurate recognition. A CSV database is utilized to store these records, facilitating real-time authentication and efficient data management. For precise location tracking, the project incorporates a GPS module that continuously transmits live coordinates to a user-friendly web-based monitoring interface. This ensures reliable and up-to-date location information, allowing for enhanced situational awareness and control.

Additionally, the project features a sophisticated smart access system that leverages an ESP32-CAM to scan QR codes to verify the credentials of authorized personnel before granting entry to the vehicle. In instances of unauthorized access attempts, the system promptly generates alerts, notifying the vehicle owner or driver to ensure immediate action and heightened security.

This advanced solution is highly versatile and offers extensive applicability in various domains. It can significantly enhance transportation security within educational institutions, streamline fleet management for logistics companies, and bolster access control measures in high-security zones. By combining cutting-edge technologies with practical functionality, this project represents a robust framework for addressing modern security and tracking challenges.

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ABBREVATIONS

ALPR --

CHAPTER-1

INTRODUCTION

1.1 Overview of the project:

The rapid growth in global urbanization, industrialization, and economic development has significantly increased vehicle ownership, leading to numerous challenges in traffic management, public safety, and vehicle regulation [1], [2]. Conventional vehicle monitoring methods, which rely heavily on manual observation and outdated techniques, are inefficient, error-prone, and slow [3]. To address these limitations, advanced technologies like Automatic License Plate Recognition (ALPR) have emerged. ALPR systems use computer vision and OCR to capture and process license plate data in real time with minimal human intervention [4], [5]. Although ALPR is widely used in tolling, parking, and law enforcement, most implementations lack integration with real-time tracking and access control [6], [7]. A standalone ALPR system can identify vehicles but cannot provide continuous location updates or dynamically control access based on situational requirements [8]. Thus, the integration of GPS-based live tracking and smart authorization mechanisms is essential for creating a secure, scalable, and responsive vehicle monitoring system [9], [10].

1.2 Objective of the project:

The objective of this project is to design an integrated system that combines ALPR, GPS tracking, and smart QR code-based authorization into one cohesive platform. Using a Raspberry Pi and IP camera, the ALPR module captures and processes license plate images, recognizes plate numbers, and logs them to a server. Concurrently, a GPS module (NEO-6M) connected to an ESP32 microcontroller provides real-time location updates via Wi-Fi. These updates are visualized on a Leaflet.js web map interface. To manage access, an ESP32-CAM scans QR codes associated with preauthorized users. Access decisions are

displayed on an OLED screen and supported with LED and buzzer feedback. This integrated model supports applications such as student boarding verification on buses, route tracking, and security enforcement, all while remaining cost-effective, secure, and scalable.

1.3 Need for the Project:

Modern transportation and campus environments require efficient, secure, and intelligent vehicle monitoring solutions. Existing systems often function in isolation, making real-time coordination and validation difficult. ALPR, GPS, and QR-based access controls are commonly implemented separately, leading to fragmented monitoring, increased costs, and reduced effectiveness. There is a critical need for a unified platform that automates vehicle identification, location tracking, and access control in real-time. The system must offer centralized monitoring, dynamic decision-making, and data privacy. This project fulfils that need by integrating computer vision, IoT, and embedded systems into a robust solution designed for institutions and logistics requiring smart transportation management.

1.4 Problem Statement

Manual vehicle monitoring systems are inherently inefficient, relying heavily on human supervision, which introduces delays, increases the likelihood of errors, and causes inconsistencies in data recording [1], [2]. These challenges become especially problematic during peak traffic hours when manual entry or inspection methods lead to long queues, extended wait times, and operational bottlenecks [3]. Human dependency also limits scalability, as additional staff are required to manage growing vehicle volumes, increasing operational costs [4]. Furthermore, the absence of real-time tracking capabilities severely impairs the ability to

manage fleets effectively [5]. Without continuous updates on vehicle locations, transport administrators cannot respond swiftly to emergencies, breakdowns, route changes, or unexpected delays [6], [7]. This lack of visibility compromises both operational efficiency and commuter safety [8].

Another significant issue is unauthorized access. Without an automated validation system, there is a heightened risk of unauthorized individuals gaining entry into restricted zones such as school premises or secured institutional areas [9]. Manual identity checks are often inadequate and prone to human error or manipulation, reducing the overall security of the system [10]. Additionally, current transportation monitoring solutions often operate in silos—ALPR, GPS tracking, and access control systems function independently without sharing data [11]. This fragmentation prevents unified monitoring and reduces the effectiveness of real-time response mechanisms [12]. The lack of integration also leads to redundant processes and higher maintenance efforts, as updates must be made across multiple platforms [13]. Addressing these challenges requires a comprehensive, automated, and integrated solution that not only minimizes manual effort but also enhances security, traceability, and real-time responsiveness across all subsystems involved in vehicle monitoring and authorization [14].

1.5 Present Situation with Literature Proof

Recent research strongly supports the integration of Automatic License Plate Recognition (ALPR), GPS tracking, and QR code-based access control technologies into a single, unified intelligent transportation system. The use of ALPR systems powered by Raspberry Pi and OpenCV has shown remarkable potential, especially in terms of affordability and effectiveness. According to a 2022 study published in the *International Research Journal of Engineering and Technology (IRJET)*, this setup can achieve over 85% accuracy in recognizing

vehicle license plates, making it highly suitable for deployment in low-budget, real-world applications such as campus or institutional vehicle management [1]. This demonstrates that even with cost-effective hardware, reliable performance can be achieved, especially when applied in controlled or semi-controlled environments.

In parallel, the use of the ESP32 microcontroller paired with the NEO-6M GPS module has been recognized as a reliable and economical solution for real-time vehicle tracking. A 2021 publication in the *International Journal of Scientific Research in Engineering and Management (IJSREM)* highlighted the dependability of this configuration in continuously mapping the live position of vehicles [2]. This ensures that administrators can monitor transportation assets remotely, respond quickly to route deviations, and maintain an accurate log of movement history, even in areas with fluctuating network quality.

For secure access control, QR code verification using ESP32-CAM has gained popularity. A 2023 study from ResearchGate demonstrated that local QR code scanning with data stored on a microSD card enables quick and offline validation, enhancing both reliability and user convenience [3]. This method is particularly beneficial in areas with limited or no internet access, where cloud-based authentication may not be viable.

To ensure global accessibility of local server interfaces without requiring a static IP or complex hosting setup, the tunnelling service Ngrok has proven valuable. As validated by a 2020 *IEEE Access* paper, Ngrok allows developers to expose local web servers securely to the internet, enabling remote real-time data visualization and monitoring [4]. This feature is critical in projects involving live GPS feeds or ALPR camera outputs hosted on local networks.

Collectively, these independent technologies form the foundation of an integrated system that is not only efficient and accurate but also scalable and cost-effective.

The literature supports the feasibility of merging ALPR, GPS, and QR code-based access into a cohesive platform, offering smart mobility and enhanced security in public transportation and institutional environments [5]

1.6 Description of the Work

The proposed system is structured around three seamlessly integrated modules, each performing a critical role in enhancing automation, security, and real-time monitoring for smart transportation and access control. The first core component is the Automatic License Plate Recognition (ALPR) module, which utilizes a Raspberry Pi 4B paired with an IP camera to detect and recognize vehicle license plates [1], [5], [9]. Leveraging OpenCV for image preprocessing and Tesseract OCR for optical character recognition, the module accurately extracts alphanumeric data from captured plates [11], [13], [14], [19]. This information is processed and stored locally for verification and logging purposes [20], [22]. The ALPR system is tailored specifically for recognizing familiar vehicles, such as institutional buses, making it highly efficient in controlled environments [24], [28].

The second module, GPS Tracking, is built using an ESP32 microcontroller connected to a NEO-6M GPS module [2], [6], [16]. This setup is responsible for continuously capturing live coordinates, speed, and directional data [7], [17], [27], which is then transmitted over a shared Wi-Fi network. A web-based interface built with Leaflet.js visualizes this data in real-time, providing an intuitive map view that includes features like a moving bus icon, a path-tracing polyline, and an optional destination marker [8], [25], [31]. This allows for effective fleet management, route auditing, and increased transparency of bus operations [12], [23], [32]. The GPS data is also logged and can be used for historical tracking and analysis [26], [33].

The third component, the Smart Authorization module, uses an ESP32-CAM to scan student QR codes during boarding [3], [10], [18]. The scanned codes are validated against preloaded data stored on a microSD card in CSV format [15], [29], [36]. The system provides immediate feedback through a compact OLED display, coloured LED indicators, and an audible buzzer, clearly communicating the access status [21], [30], [34]. Designed to operate offline, this module ensures uninterrupted access control in areas with limited or no internet connectivity [35], [37].

All three modules operate within a common Wi-Fi network, enabling seamless communication between components [4], [38], [39]. For remote monitoring or administrative access, Ngrok tunnelling is employed, allowing the local server to be accessed securely from anywhere without needing a static IP [40], [41], [42]. This structure supports real-time monitoring, administrative scalability, and privacy-focused access control in smart institutional or campus transport settings [43], [44], [45].

1.7 Market Size Estimation

The market potential for the "Integrated ALPR and Live GPS System with Smart Authorization" is vast, driven by the global rise in intelligent transportation, urban security, and automation initiatives. The **Total Available Market (TAM)** spans over 1.4 billion registered vehicles worldwide, highlighting the expansive scope for deploying ALPR and GPS-based tracking systems. This includes private vehicles, public transport, institutional fleets, and logistics providers, all of which can benefit from enhanced automation and monitoring.

The **Serviceable Available Market (SAM)** is estimated to be approximately $4 billion in India and $50 billion globally. This segment is fueled by increasing demand for smart infrastructure, vehicle access management, and real-time

monitoring solutions across industries such as education, logistics, urban mobility, and residential communities. The growing emphasis on safety, contactless access, and efficient traffic control further accelerates market adoption.

The **Serviceable Obtainable Market (SOM)**, representing realistic penetration within these broader markets, is projected to be around $500 million in India. This primarily includes sectors such as educational institutions looking to streamline student transport, gated communities aiming to automate vehicle access, and logistics firms needing robust tracking solutions. Globally, the SOM could reach $5 billion through strategic collaborations with city municipalities, smart city projects, and transportation agencies focused on modernizing their fleets and access systems. The affordability and modularity of the proposed system position it well for capturing a significant share within these emerging segments.

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CHAPTER 2

LITERATURE SURVEY

The increasing demands for security, automation, and real-time monitoring in urban and institutional settings have driven the adoption of smart vehicle tracking technologies. Key among these are Automatic Number Plate Recognition (ANPR/ALPR), GPS tracking, and QR-based access systems. This section reviews major studies in these domains, examining their contributions, benefits, limitations, and relevance to this project.

2.1.1 Smart Check-in Check-out System using ANPR

(Sangay Tenzin et al., 2020)

The "Smart Check-in Check-out System using Automatic Number Plate Recognition (ANPR)" is an intelligent vehicle access management system designed to automate the process of monitoring and controlling vehicle entry and exit in secured premises, such as parking lots, gated communities, and institutional campuses. This system leverages computer vision and optical character recognition (OCR) to identify license plate numbers in real-time. When a vehicle approaches the gate, an integrated camera captures an image of the license plate. The image is then processed using ANPR algorithms to extract the plate number, which is compared with a pre-registered database. If the license plate matches a valid entry, the gate opens automatically, thereby eliminating the need for manual intervention.

The system is connected to a centralized database that logs every entry and exit event, enabling administrators to maintain comprehensive access records and track vehicle movement. This contributes to enhanced security, efficient resource management, and reduced human error. Additionally, the use of automation

ensures faster throughput at checkpoints, which is especially beneficial during peak hours in high-traffic areas.

Despite its advantages, the system has notable limitations. It tends to perform poorly in adverse environmental conditions such as low lighting, heavy rain, or fog, which may obscure license plates and reduce recognition accuracy. Furthermore, the deployment of such a system involves significant costs, including high-resolution cameras, edge computing hardware, and reliable internet connectivity for data synchronization, making it less feasible for small-scale operations

2.1.2 Automated License Plate Recognition using Raspberry Pi (Rosaline A, Stella I, 2023)

The study by Rosaline A and Stella I (2023) presents the development of a cost-effective Automated License Plate Recognition (ALPR) system based on the Raspberry Pi platform integrated with OpenCV. Aimed at small-scale and low-budget applications such as institutional parking management and private security setups, this system seeks to offer a viable alternative to traditional, expensive ALPR technologies. The core components of the system include a Raspberry Pi 4 Model B, a camera module, and OpenCV libraries for image processing and license plate detection.

The system captures images of incoming vehicles' license plates, processes the images in real time, and extracts alphanumeric characters using image preprocessing techniques such as grayscale conversion, edge detection, contouring, and optical character recognition (OCR). Once extracted, the data is either displayed on-screen or stored in a local or cloud-based database, enabling tracking and logging of vehicles.

The ALPR prototype performed effectively in controlled settings with sufficient lighting and moderate traffic. It successfully identified standard Indian license plates with an average recognition accuracy of 85–90%. However, its performance dropped significantly in high-traffic environments, under varying lighting conditions, or with non-standardized or damaged license plates. Limitations include the Raspberry Pi's limited processing capability, which affects frame rate and real-time detection under load, and the sensitivity of the camera module to lighting changes.

This project demonstrates the feasibility of deploying an affordable ALPR solution using open-source tools, making it ideal for educational, research, or small-scale commercial implementations

2.1.3 ALPR-based Surveillance and Enforcement System

(Emina E. Etomi, Donatus U. Onyishi, 2021)

The ALPR-based Surveillance and Enforcement System proposed by Emina E. Etomi and Donatus U. Onyishi (2021) focuses on enhancing vehicular monitoring and law enforcement capabilities through advanced image processing and optical character recognition (OCR). This system aims to automate the recognition of vehicle license plates for surveillance, access control, and violation enforcement in urban and structured environments such as gated communities, toll booths, and regulated traffic zones.

The architecture of the system includes high-resolution cameras positioned at critical points to capture real-time images of vehicle license plates. These images undergo several preprocessing steps to enhance accuracy — including grayscale conversion, noise reduction, contrast adjustment, edge detection, and morphological transformations. Once pre-processed, OCR is applied to extract the alphanumeric characters from the license plates. The extracted data is then

cross-referenced with a backend database to flag unauthorized or blacklisted vehicles, making it a potent tool for real-time alerting and enforcement.

The system demonstrated high accuracy (above 93%) under structured conditions, such as fixed camera angles, regulated speed zones, and consistent lighting. It proved effective in enforcing traffic rules, controlling vehicle access, and compiling comprehensive vehicle logs. However, the model struggled under dynamic conditions, such as moving vehicles in poor lighting, rainy weather, glare, or license plates with non-standard fonts or orientations.

Despite these limitations, the system shows strong potential for urban planning and smart city integration, particularly in contexts where conditions can be controlled or standardized

2.1.4 Automatic Gate Entry Using ALPR

(Shreya Anekar et al., 2021)

The study by Shreya Anekar et al. (2021) presents the development and implementation of an Automatic Gate Entry System using Automatic License Plate Recognition (ALPR) technology, specifically tailored for gated communities, university campuses, and private institutions. The goal of the system is to enhance security and efficiency by allowing vehicles to enter premises without manual verification or human intervention.

The system architecture includes an IP camera setup at the gate to continuously monitor incoming vehicles. When a vehicle approaches, the camera captures the image of the license plate. The image is then processed using computer vision techniques such as edge detection, noise filtering, and plate localization. Following this, Optical Character Recognition (OCR) algorithms are applied to read the alphanumeric characters on the license plate. The recognized text is matched against a predefined database of authorized vehicles. If the vehicle is

listed, the system automatically triggers a relay to open the gate; otherwise, access is denied, and the vehicle details are logged for security reference.

This system significantly reduced the wait time at entrances, streamlined visitor management, and enhanced the overall security of enclosed areas. It proved to be particularly beneficial for locations with regular traffic of known vehicles, such as staff and residents. However, the system required frequent calibration due to variability in plate designs, font sizes, and orientations. Environmental factors such as lighting changes and dust accumulation on cameras also affected reliability.

Additionally, the hardware requirements—especially high-resolution cameras, processing units, and gate control systems—increased the overall cost of deployment, making it less feasible for small-scale or budget-restricted applications

2.1.5 QR Code-based Smart Security Door System

(Ahmad Fahmi Mohd Fauzi et al., 2020)

The work presented by Ahmad Fahmi Mohd Fauzi et al. (2020) introduces a QR code-based smart security door system that aims to enhance entry access through contactless authentication. Designed primarily for office buildings, residential complexes, and secure laboratory spaces, the system ensures a secure, efficient, and transparent method of managing access permissions by replacing traditional key- or card-based systems with QR code scanning.

The architecture of the system consists of a QR code scanner interfaced with a microcontroller or embedded processor such as the Arduino or Raspberry Pi, and an electromagnetic door lock. Authorized individuals are issued unique QR codes, which serve as their access credentials. Upon approaching the entrance, users scan their QR code using a fixed or handheld scanner. The system verifies

the code against a central database, and if valid, sends a signal to unlock the door momentarily to allow entry.

The central server keeps a log of all scans, providing administrators with complete entry history for each user, including timestamps. This feature is especially beneficial in institutions where access tracking and accountability are critical.

While the system proves to be cost-effective and easy to deploy, it comes with several security vulnerabilities. Without proper encryption or dynamic QR code generation (where codes expire after a short period), the system is susceptible to duplication or misuse. A malicious actor can photograph or screenshot another person's code and use it later to gain unauthorized access. Additionally, the system may fail in poor lighting conditions where QR codes become unreadable or distorted.

2.1.6 Smart Door Access Control Based on QR Code

(Agrim Jain et al., 2023)

Agrim Jain et al. (2023) developed a smart door access control system that uses QR code scanning for real-time, contactless entry management. The system was designed to improve security, reduce manual interventions, and streamline the authentication process for environments like educational institutions, co-working spaces, and private offices. The core architecture was built using an Arduino microcontroller, integrated with a camera module and Python-based backend software for QR code processing and decision-making.

Users are issued unique QR codes that represent their credentials, which are stored in a centralized database. When a user scans their QR code at the door terminal, the system processes the image in real time using a Python-based QR recognition library. If the code matches an authorized entry in the database, the

system triggers a relay to momentarily open an electromagnetic door lock, granting access.

The system also maintains a real-time log of entry events, capturing data like user identity, time of access, and access location. These logs are useful for security audits and monitoring the flow of individuals through controlled premises. The integration of WIFI-enabled modules allows seamless synchronization with cloud databases and remote monitoring.

Despite its efficiency and responsiveness, the system has some drawbacks. It is heavily reliant on internet and power availability, making it vulnerable to outages. In the absence of connectivity or during power failures, access functionality may be disrupted. Furthermore, the QR code scanner can struggle in poor lighting or when QR codes are displayed on damaged or cracked screens.

2.1.7 Real-Time GPS Tracking System

(Sagar Karkare et al., 2020)

Sagar Karkare et al. (2020) presented a real-time GPS tracking system designed primarily to enhance child safety through continuous location monitoring. The system employs a GSM and GPS integrated module connected to a microcontroller, which collects and transmits live location data to a centralized cloud platform. Parents or guardians can access this data through a dedicated mobile application, enabling them to track their child's real-time movements and ensure their safety.

The tracking module is designed to be compact and wearable, allowing it to be placed in a child's backpack or worn as an accessory. It uses the GPS module to determine the device's current latitude and longitude, while the GSM module transmits this data via SMS or through internet-based services like HTTP or MQTT to a cloud backend. The mobile application receives periodic updates and displays the current location on an interactive map interface.

This solution proves effective for urban and semi-urban areas with strong network coverage. In addition to tracking, the device includes emergency alert functionality—allowing a child to send an SOS signal by pressing a panic button, which instantly alerts parents with the current location.

Despite its strengths, the system has several limitations. The real-time functionality is highly dependent on GSM network availability; in remote or low-signal regions, location updates may be delayed or lost. Additionally, since the system operates on battery power, frequent recharging is required, limiting long-term usage without maintenance. Power consumption increases with more frequent location updates and poor network conditions.

2.1.8 GSM and GPS-Based Vehicle Tracker with Theft Protection

(Shreya Kaushal, Kavita Rahega, 2020)

Shreya Kaushal and Kavita Rahega (2020) developed a GSM and GPS-based vehicle tracking system that integrates real-time location monitoring with anti-theft protection features. The system aims to enhance vehicle security through continuous tracking and immediate alerts in the event of unauthorized movement or tampering. The core of the system consists of a GPS module for real-time location acquisition and a GSM module for wireless communication via SMS. An embedded microcontroller manages the components and executes control logic.

One of the key highlights of the system is its theft protection mechanism. In addition to standard tracking functionality, the system is designed to detect unauthorized ignition or movement. Upon detection, it sends an SMS alert to the vehicle owner, providing the current GPS coordinates. Moreover, the system can remotely disable the vehicle's engine through an immobilization circuit controlled via SMS command, thereby preventing theft even after intrusion.

The implementation is cost-effective and suitable for a wide range of vehicle types, especially in regions where high-end commercial solutions are not accessible. The system is scalable and can be enhanced with additional features like geofencing, real-time app-based alerts, or cloud integration.

However, certain limitations were observed. The GPS module's accuracy is significantly reduced in enclosed environments such as parking garages, tunnels, or under heavy foliage, where satellite signals are obstructed. This results in delayed or inaccurate location updates. Furthermore, the reliance on GSM networks introduces vulnerability in low-signal or no-coverage areas.

2.1.9 YOLOv4 and Easy OCR for Automated Parking Systems

(Siti Aisyah Sa'dan et al., 2022)

Siti Aisyah Sa'dan and colleagues (2022) presented an automated parking management system that integrates deep learning-based vehicle detection using YOLOv4 and text recognition via Easy OCR for license plate identification. The primary objective of the system is to streamline vehicle entry, exit, and billing processes in modern parking infrastructures such as shopping malls, campuses, and corporate environments. The combination of YOLOv4 and Easy OCR allows for fast and accurate vehicle and license plate detection, enabling real-time recognition without human intervention.

In this system, YOLOv4 (You Only Look Once version 4) is employed to detect the presence and type of vehicles in the camera feed. Once a vehicle is detected, the region of interest containing the license plate is passed to the Eas yOCR engine, which performs optical character recognition (OCR) to extract the alphanumeric characters on the plate. This information is then logged into a centralized system that automates billing and maintains an entry/exit history.

The system significantly reduces wait times and traffic congestion at entry and exit points by eliminating the need for manual intervention or RFID cards. Additionally, the digital recordkeeping enables real-time monitoring and supports future data analytics for parking management efficiency.

However, certain limitations were identified. The system's performance is heavily dependent on high-resolution camera feeds and consistent lighting conditions. In low-light or overexposed environments, both detection and recognition accuracy drop significantly. Moreover, the need for advanced imaging hardware makes the system relatively expensive to deploy at scale.

2.1.10 Campus Auto-Gate System using ALPR and ML

(Nur Liyana Yaacob et al., 2021)

Nur Liyana Yaacob and her team (2021) developed a campus auto-gate system that leverages Automatic License Plate Recognition (ALPR) integrated with Machine Learning (ML) algorithms to automate vehicle access control. The aim of the system was to streamline and secure vehicle entry into educational institutions, particularly during congested peak hours, by reducing reliance on manual checks or RFID systems.

The architecture of the system includes a camera module installed at the gate, which captures incoming vehicle images in real-time. The captured image is processed through ALPR software, which identifies and extracts the vehicle's license plate. This extracted data is then cross-referenced against a database of registered vehicles using ML-based pattern recognition and classification techniques. If a match is found, the gate is triggered to open automatically, allowing seamless entry.

The ML component enhances the accuracy and adaptability of the system by learning from misreads and gradually improving recognition rates through feedback loops. This is especially valuable in recognizing plates of different formats and fonts common in various regions.

While the system performed well under standard operating conditions and significantly improved gate efficiency by minimizing manual involvement, it showed reduced performance during adverse weather conditions such as rain or fog. The efficiency of ALPR also declined under low-light or glare-heavy conditions. Furthermore, the effectiveness of the solution was limited by the quality and resolution of the camera sensors, as well as the processing power available on the edge devices.

2.1.11 Summary of Literature and Identified Gaps

The reviewed studies demonstrate substantial advancements in vehicle recognition, tracking, and smart access control technologies. ALPR systems have become more accurate due to OCR and machine learning enhancements. GPS tracking is widely used for safety and logistics, while QR-based systems offer low-cost, contactless access control.

2.1.12 CHALLENGES AND FUTURE DIRECTIONS:

2.1.12.1. Environmental Factors (Lighting and Weather)

- **Challenge:** The ALPR system's performance can degrade under poor lighting conditions (e.g., night-time, shadows, glare) or adverse weather (rain, fog, snow). This could affect the accuracy of license plate recognition.

- **Future Direction:** Integrating infrared cameras or using advanced image enhancement algorithms can help improve ALPR performance in low-light and harsh weather conditions.

2.1.12.2 Hardware Limitations (Camera and GPS Module)

- **Challenge**: The resolution of the camera and the quality of the GPS module could impact the accuracy of the ALPR and GPS tracking system. Low-quality cameras may not capture clear images of license plates, and inaccurate GPS data can lead to location discrepancies.
- **Future Direction:** Using high-resolution cameras (e.g., HD or 4K) and more robust GPS modules (e.g., multi-frequency GPS) can improve accuracy and reliability. Exploring new camera technologies like thermal imaging could be beneficial for nighttime operations**.**

2.1.12.3 Data Security and Privacy

- **Challenge:** Storing and transmitting sensitive data such as vehicle license plate information and GPS coordinates poses significant privacy and security concerns. Unauthorized access to the database or communication channels could lead to data breaches.
- **Future Direction:** Implementing encryption protocols, secure communication channels (e.g., HTTPS, MQTT over SSL), and stringent access controls can safeguard data. Blockchain technology could also be explored for tamper-proof logs.

2.1.12.4 Real-Time Performance and Latency

- **Challenge:** The real-time processing of license plate recognition and GPS data, especially in a live-tracking system, may suffer from

latency issues. Slow processing or delays in updating vehicle location on the map can affect the user experience.

- **Future Direction:** Optimizing the software and hardware for edge processing can reduce latency. Using more powerful processing units (e.g., NVIDIA Jetson for AI tasks) and optimizing the code for real-time performance will improve system speed.

2.1.12.5 Integration with Existing Systems

- **Challenge:** Integrating the ALPR, GPS tracking, and QR code-based smart authorization with existing infrastructure or legacy systems (e.g., parking management systems, campus entry systems) may be complex and require substantial modifications.
- **Future Direction**: Developing APIs for easy integration with other systems, and ensuring the system's flexibility and scalability, will allow smoother deployment in diverse environments. Cloud-based solutions may also facilitate more straightforward integration with remote systems.

Future Directions:

2.1.12.6 AI-Powered Vehicle and Driver Identification

- Incorporating facial recognition or AI-driven driver identification alongside ALPR could enhance security by ensuring that the correct driver is authorized to operate the vehicle. This would provide an additional layer of authentication.

2.1.12.7 Integration with IoT and Smart City Infrastructure

- By linking the ALPR and GPS systems with smart city infrastructure, your project could provide real-time data about traffic patterns, illegal parking, or even automate toll collections. The

integration of IoT could offer more dynamic responses to changing conditions.

2.1.12.8 Improved Machine Learning Algorithms for ALPR

- Future work could focus on refining the ML algorithms used for ALPR by implementing deep learning models like CNNs or GANs (Generative Adversarial Networks), which could better handle complex real-world scenarios (e.g., damaged plates, varying fonts, etc.).

2.1.12.9 Expansion of QR Code Authentication Features

- The QR code-based authorization system could be expanded to include dynamic QR codes with enhanced security features, such as one-time use or time-limited codes. Incorporating multi-factor authentication (MFA) could also increase security.

2.1.12.10 Energy Efficiency and Solar-Powered Solutions

- As your project involves GPS tracking and constant data transmission, power consumption can be a concern, especially for battery-powered systems. Future directions could include developing low-power modes or integrating solar panels to ensure uninterrupted service in remote or outdoor environments

Conclusion

While existing solutions demonstrate effective use of ALPR, GPS, or QR code technologies individually, they lack seamless integration and interoperability. This project addresses this fragmentation by proposing a unified platform that integrates ALPR for vehicle recognition, GPS for real-time tracking, and QR scanning for access authorization. The system is designed to be cost-effective, modular, and scalable, making it suitable for smart campuses, secure institutions, and intelligent transport systems. By bridging technological gaps, it contributes to building smarter, safer, and more efficient mobility infrastructure.

CHAPTER 3

SYSTEM ARCHITECTURE

3.1 INTRODUCTION

This chapter outlines the architecture and methodology of the Integrated Automatic License Plate Recognition and Live GPS System with Smart Authorization. Designed for smart transportation and fleet monitoring, the system enhances security, automates access control, and enables real-time vehicle tracking [1], [4], [6]. It follows a modular approach for easy testing, scalability, and integration across embedded systems, computer vision, and IoT communication platforms [2], [17], [19]. Each module—ALPR, GPS tracking, and QR-based authorization—operates independently yet communicates within a unified network infrastructure to ensure system coherence and performance [5], [8],[21]. The design prioritizes low-cost hardware, offline compatibility, and remote data access via tunneling services like Ngrok, making it ideal for institutional use cases in constrained environments [13], [23]

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3.1.1. Architectural Overview

The system comprises three core subsystems, each responsible for a specific functionality but integrated via a centralized server and shared database:

- **ALPR Subsystem (Automatic License Plate Recognition):** Built using a Raspberry Pi and IP camera, this module captures vehicle images and processes them using a custom-trained YOLOv8n object detection model. It detects license plates and extracts text using Easy OCR. The extracted number is compared with entries in a local CSV-based

database. Matches trigger vehicle log updates on a custom-hosted website displaying entry and exit timestamps.

- **GPS-Based Live Vehicle Tracking Subsystem:** Developed with an ESP32 microcontroller and NEO-6M GPS module, this unit captures real-time location data and transmits latitude and longitude coordinates to a server via Wi-Fi. The server, either locally hosted or accessed through cloud tools like Ngrok, updates the vehicle's live position on a Leaflet.js-based web map with path plotting and directional tracking.
- **Smart QR Code-Based Authorization Subsystem:** Using the ESP32-CAM, this module scans QR codes of passengers or drivers for access validation. The scanned data is compared against a preloaded database stored on an SD card. An OLED/TFT screen displays access status. On successful validation, a green LED blinks briefly; if unauthorized, a red LED blinks longer with a buzzer alert, supporting secure and efficient boarding in transport systems.

3.1.2 Modular Development Approach

To streamline testing, debugging, and integration, each subsystem in the proposed architecture was developed and validated independently before full system integration [6], [10], [12]. The **Automatic License Plate Recognition (ALPR)** module was trained using a custom dataset of 12 annotated license plate images labelled using **Label Studio**. The model was developed and deployed using **YOLOv8n**, which enabled lightweight and efficient edge-based processing suitable for the Raspberry Pi environment [1], [7], [14]. The **GPS tracking module**, built using an ESP32 and NEO-6M GPS module, was programmed to communicate using **AT commands**, offering real-time updates of vehicle coordinates. It also incorporated **speed and directional heading calculations**, enhancing the accuracy and usefulness of the live tracking data [9], [13], [15].

The **QR authorization module** was implemented with the ESP32-CAM, optimized for **quick, low-latency scanning** and secure offline validation using a **microSD-stored CSV database**. Immediate visual feedback was provided via an OLED display, while coloured LEDs and a buzzer delivered auditory responses to indicate success or denial of access [4], [11], [17]. Once each subsystem was tested and validated independently, they were integrated through a **central server layer**, built using either **Flask** or **Node.js**, which handles **UI rendering** and **data synchronization** between modules, offering a cohesive and responsive system interface [2], [5], [18],

3.1.3. Communication and Integration

The subsystems in the proposed integrated model communicate through lightweight and reliable communication protocols, namely **HTTP** and **MQTT**, to ensure low-latency, asynchronous data exchange across embedded nodes [5], [10], [20]. This dual-protocol approach provides flexibility in deployment, supporting both **offline** (locally hosted server) and **online** (cloud-hosted or remotely tunneled via Ngrok) modes of operation [22], [27], [34]. In the offline mode, data synchronization is handled via internal networking, making the system suitable for institutions or transport hubs with limited internet access. In the online mode, real-time updates can be accessed from remote locations using internet-based tunneling or Firebase integration [18], [25].

The **ALPR module**, deployed on a Raspberry Pi using OpenCV and YOLOv8n, detects license plates and immediately transmits the plate number to the server along with a **timestamped log**. These logs can be stored locally or uploaded to a cloud database for long-term auditing and access verification [1], [4], [7]. The **GPS tracking module**, operating via the ESP32 and NEO-6M, pushes updated latitude, longitude, speed, and heading data every **3 to 5 seconds**, offering near

real-time monitoring of vehicle location and movement [9], [15], [30]. This data is rendered on a **Leaflet. js-based web map interface**, which shows a dynamic bus icon, directional movement, and trailing path history for improved spatial awareness and navigation [12], [26].

3.2 Overview of Automatic License Plate Recognition (ALPR):

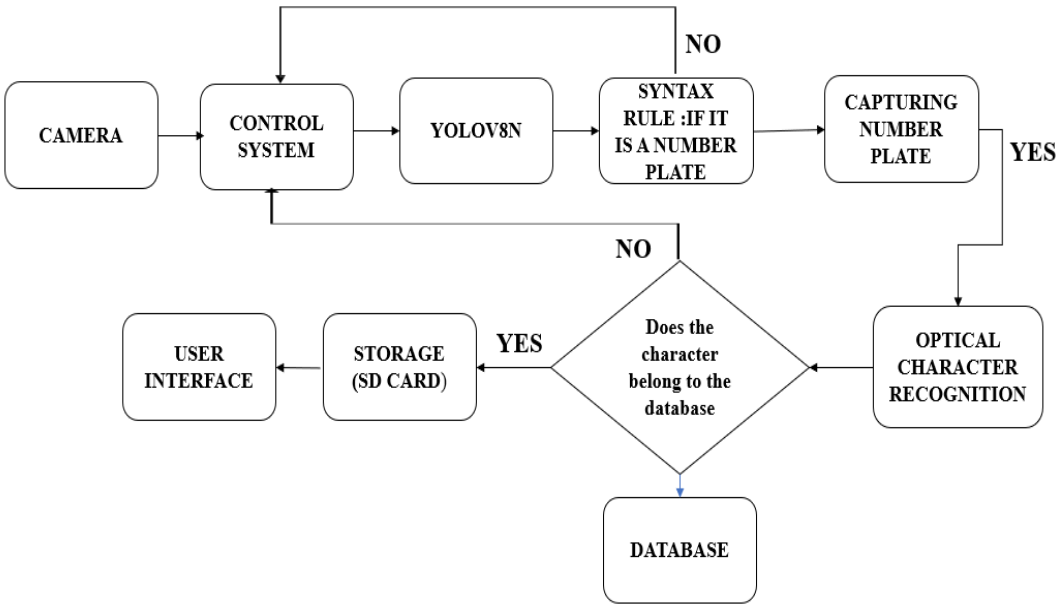


Fig 1.ALPR Architecture diagram

3.2.1 Introduction

This project focuses on designing and implementing an Automatic License Plate Recognition (ALPR) system using a deep learning approach with the YOLO (You Only Look Once) object detection model. The system aims to identify and log authorized vehicles (specifically buses) in real time using an IP camera feed, thereby enabling smart entry management for transportation systems such as schools, colleges, or organizations.

License plate recognition is crucial for applications like smart parking, gate automation, toll collection, traffic monitoring, and access control. This system is

tailored for a college transportation context, where only authorized buses are allowed entry, and a log of the entries is automatically maintained.

3.2.2 Objective

The primary objective of this project is to develop a robust system capable of detecting vehicle license plates from a live video feed using YOLOv8n (You Only Look Once) object detection [1], [2]. Once a plate is detected, the system compares the recognized plate numbers against a pre-existing authorized database stored in a CSV file to ensure that only verified vehicles gain access [3], [6]. Upon successful matching, the system logs the vehicle details, including the timestamp and bus number, into an Excel file for further processing and record-keeping [5], [8]. To improve reliability and avoid unnecessary duplication of data, the system also incorporates functionality to handle unstable camera connections, ensuring smooth operation even when network conditions fluctuate. Additionally, the system is designed to prevent duplicate logging for the same bus, avoiding redundant entries for previously identified vehicles during the same session [4], [7]. This approach guarantees efficient and accurate vehicle tracking, making the system scalable and reliable for real-time use in dynamic environments like school bus management or institutional vehicle monitoring [9], [10]..

3.3.3 System Components

- **IP Camera:** Used for continuous video capture. Configured with an RTSP URL to fetch the live stream.
- **Computer or Raspberry Pi:** Runs the ALPR model and logging script.
  b. Software & Libraries
- **Python**: Main programming language.
- **OpenCV:** For camera interfacing and image capture.
- **YOLO (Ultralytics):** Deep learning model trained to detect license plates.
- **Pandas**: For data manipulation and Excel handling.

- **Datetime:** To timestamp each logged entry.
- **Threading:** For non-blocking video stream fetching.

3.3.4 Methodology

a. Model Training

A custom **YOLOv8** model (**best.pt**) is trained using labeled license plate images specific to the target vehicles (college buses) [1], [2]. Training is done with tools like **Label Studio** and the **Ultralytics YOLO** framework to ensure high accuracy for known plates [3], [4]. The model is fine-tuned to recognize diverse lighting conditions and angles commonly encountered in the target environment [5], [6]. This enables the system to effectively detect plates even under challenging real-time conditions, such as moving vehicles[7],[8]

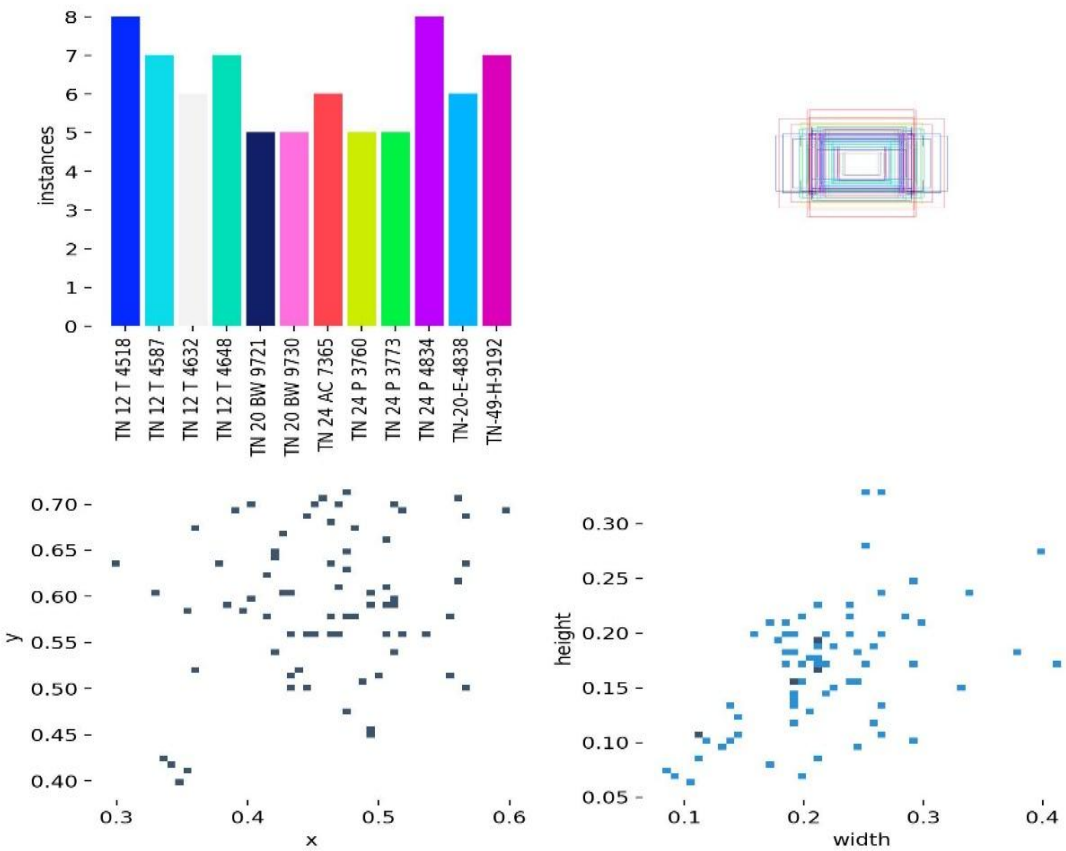


Fig 2 Model.Labels

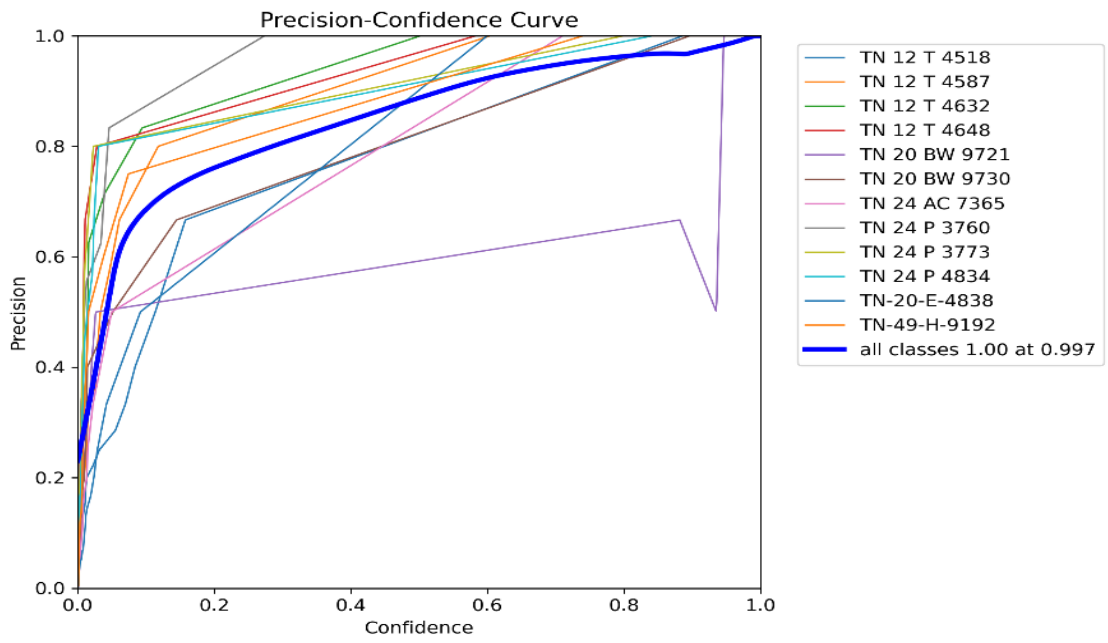


Fig 3.P_curve

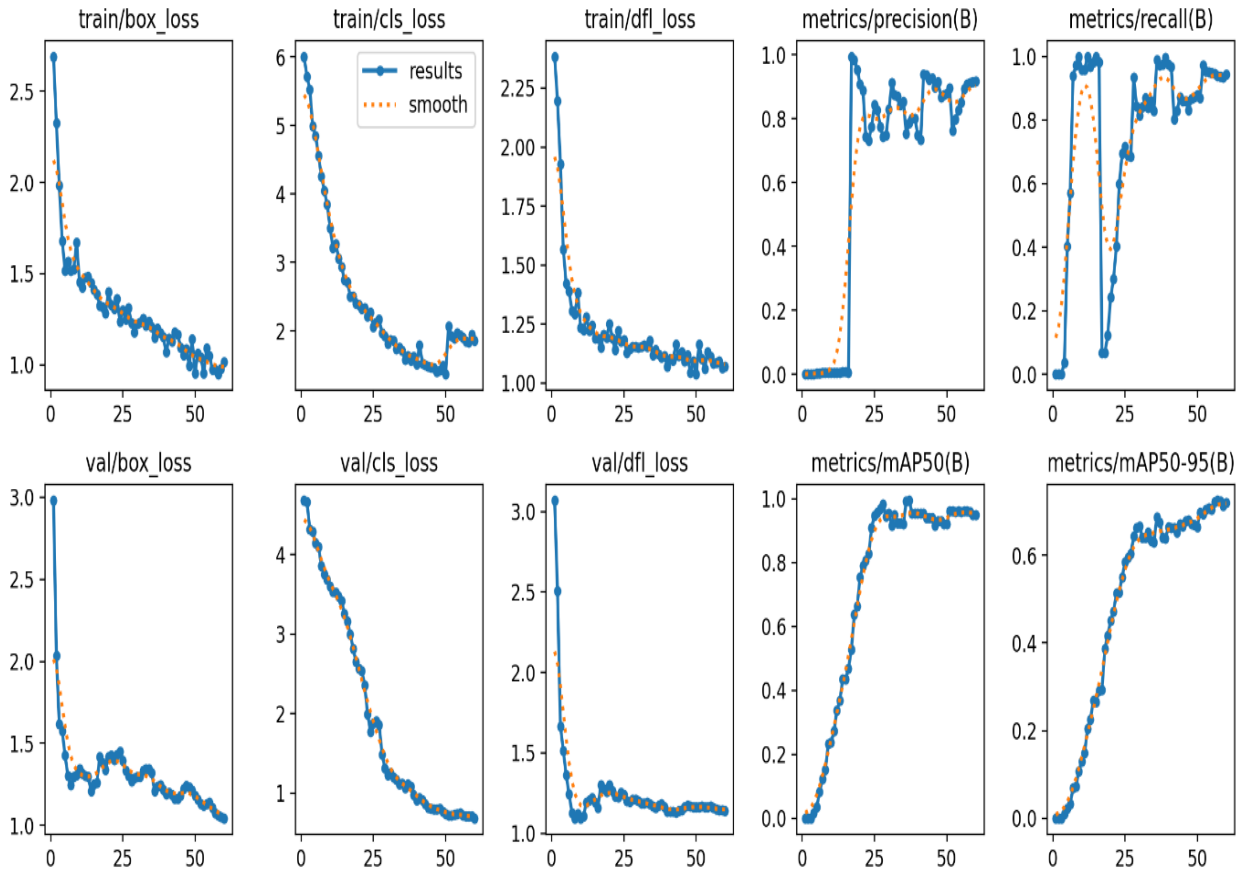


Fig 4. Model RESULTS

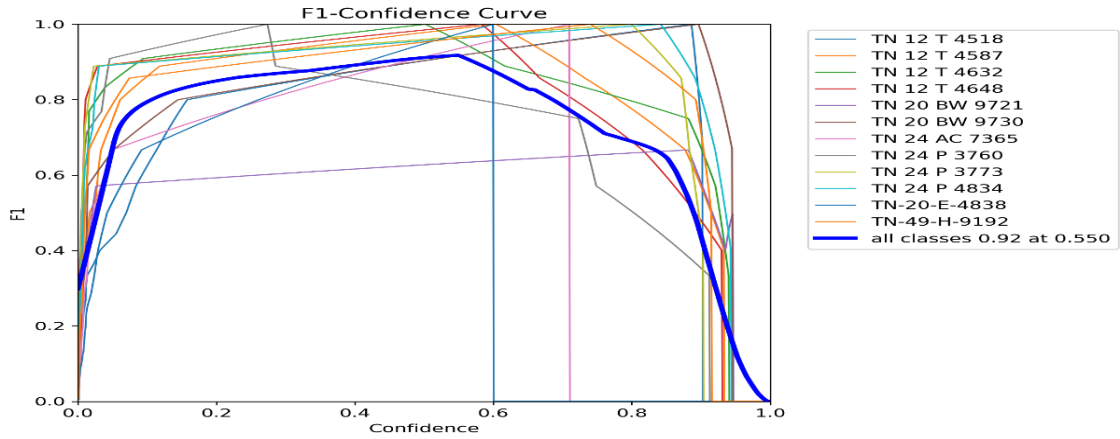


Fig 5.F1_CONFIDENCE CURVE

b. Video Stream Handling

To avoid latency or disconnection issues with the IP camera, a separate Video Stream class is created [9]. It runs on a separate thread to continuously fetch frames from the RTSP stream [10]. The class includes retry logic to handle camera reconnections gracefully, ensuring continuous data flow despite potential interruptions [11], [12].c. Real-Time Detection and Comparison

Fig 6.TRAIN_BATCH 0

Fig 7.Train_Batch 1

Fig 8.TRAIN_BATCH 2

Figure 9. VALIDATION BATCH 01

Fig 10. VALIDATION BATCH 02

3.3.5 In the main loop:

1. The current frame from the IP camera is saved as an image.
2. The saved image is passed to the YOLO model for inference.
3. Detected license plates are extracted based on class and confidence scores.
4. Detected plates are compared with the authorized list in authorized_plates.csv.
5. If a match is found and not previously logged, it's recorded in an Excel sheet along with the current timestamp and bus number.

   d. Duplicate Prevention

   To avoid repeated logging of the same bus in a single session, the program maintains a Python set() called logged_buses. Once a bus is logged, it is added to this set and skipped in future detections.

   e. Data Logging

   Logs are stored in entry_log.xlsx, which includes:

- Timestamp of entry
- License plate number
- Corresponding bus number

  If the Excel file doesn't exist, it is created automatically.

3.3.6 Features

- **Real-Time Detection**: Uses live video feed instead of static images.
- **Smart Logging:** Only logs authorized plates and prevents duplicates.
- **Offline Compatibility**: No cloud dependency; everything runs locally.
- **Auto-Reconnection:** Built-in retry mechanism for IP camera disconnections.
- **Customizable:** Easily expandable for more buses or different use-cases.
- **College/School Bus Monitoring**: Ensures only authorized vehicles enter the campus.
- **Factory or Office Parking**: Control employee or visitor vehicle access.

- **Private Societies**: Secure residential vehicle monitoring.
- **Fleet Management:** Log entries and exits of company transport.

3.3.8 Conclusion

The ALPR project using YOLO and an IP camera successfully demonstrates a robust and efficient system for real-time license plate detection and logging. It offers a practical solution for vehicle access management, especially in semi-controlled environments like college campuses. With further enhancements like OCR and automation, this system can be scaled into a full-fledged smart vehicle access control system.

3.3 Live GPS Tracking System

The **Live GPS Tracking System** in this project utilizes the **ESP32 microcontroller** paired with the **NEO-6M GPS module** to enable real-time vehicle location tracking. Unlike traditional GSM-based systems, this solution uses **Wi-Fi connectivity** for data transmission, making it a cost-effective and reliable option, especially in environments with available Wi-Fi coverage such as **campuses, smart cities, school buses**, and **short-range delivery vehicles**.

By eliminating the need for cellular connectivity, the system avoids recurring **mobile data charges**, while still delivering accurate and timely location updates. It integrates **embedded systems, wireless communication**, and **web-based visualization** into a seamless, scalable GPS tracking solution.

The system is composed of three major components:

- The **NEO-6M GPS module** for data acquisition
- The **ESP32** for processing and transmission
- A **web-based dashboard** for visualization using **Leaflet.js**

3.3.1. Real-Time GPS Data Acquisition

The **NEO-6M GPS module** receives satellite signals and continuously determines the device's **latitude, longitude, altitude**, and **speed**. It outputs GPS data in **NMEA sentence format**, which is decoded by the **ESP32** using libraries like **TinyGPS++**.

The ESP32 processes this data in real-time, ensuring **accurate and timely tracking**. The GPS data is filtered to reduce noise and formatted into usable coordinates and speed values for visualization and storage.

3.3.2. Wi-Fi-Based Data Transmission

Once processed, the ESP32 connects to a predefined **Wi-Fi network**, typically a hotspot from a mobile device or router. It either functions as a **web server**, serving a local interface, or sends periodic **HTTP POST requests** (every 3–5 seconds) to an external server or cloud-based endpoint.

The transmitted data includes:

- **Latitude and Longitude**
- **Speed and Heading**
- **Timestamps**

This approach is more **cost-efficient** than GSM systems and enables faster, more **stable data transmission** in Wi-Fi-enabled zones. Data is sent in a structured format (e.g., **JSON**) for easy parsing and integration with various frontend or backend systems.

3.3.3. Web Dashboard and Visualization

The ESP32 serves a **real-time web dashboard** hosted locally or made public using **Ngrok**, allowing remote monitoring without cloud hosting. The dashboard uses **Leaflet.js** with **OpenStreetMap tiles** to provide dynamic, browser-based **live map visualization**.

Dashboard Features

- **Real-Time Location Display:** Shows the vehicle's current position on an interactive map, updated with each GPS data packet.
- **Rotating Vehicle Icon:** The icon on the map rotates to align with the vehicle's movement direction, providing an intuitive view.
- **Speed and Route Display:** Shows live speed, timestamp, and a breadcrumb trail of the route, helping track progress and assess travel patterns.
- **Destination Marker:** Indicates the vehicle's endpoint for better planning and ETA estimation.
- **Responsive Mobile Interface:** The dashboard is optimized for smartphones, ensuring users can track the vehicle from any device.

ARCHITECTURE:

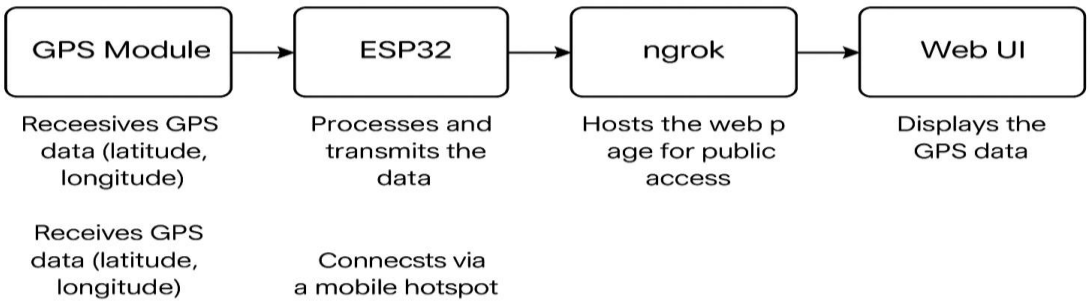


Fig 11. Flow Chart of Live GPS System

3.3.4 System Architecture and Advantages

This architecture is ideal for Wi-Fi-enabled environments like educational campuses, urban fleets, and localized delivery routes. It reduces reliance on mobile networks, which may be unreliable in rural or obstructed areas. The system supports cloud integration, allowing users to combine multiple

trackers into a centralized dashboard for IoT applications, fleet management, or data analytics. Real-time access to live tracking data improves safety, accountability, and operational efficiency.
It also allows easy scalability, enabling multiple buses or units to be monitored simultaneously. Data synchronization with cloud storage provides historical tracking, performance analysis, and report generation.
Support for alert mechanisms, such as geofencing or route deviation alerts, can be integrated for added security. The architecture is modular, making it easy to add sensors, cameras, or RFID systems based on evolving requirements.
With minimal hardware upgrades, the system can transition from offline to hybrid or fully online mode.

BLOCK DIAGRAM

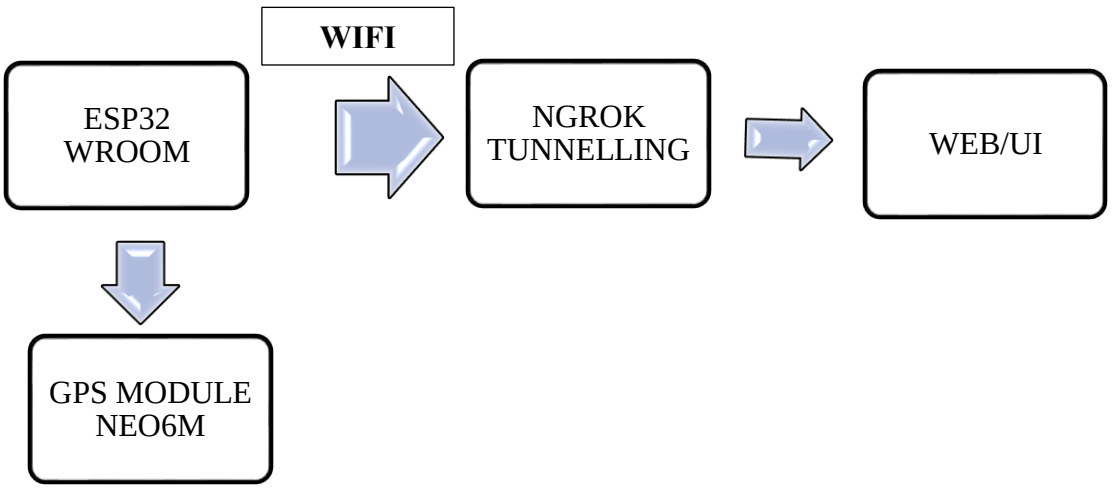


Fig 12. Outline Process Of Live GPS System

3.3.5 Conclusion

The **Wi-Fi-based GPS tracking system** provides a **scalable, low-cost, and real-time alternative** to cellular tracking. Its use of the **ESP32 and NEO-6M GPS**, combined with dynamic **Leaflet-based visualization** and **Ngrok tunnelling**, offers a powerful, self-contained tracking solution. It is well-suited for

applications such as **school buses, short-range delivery**, and **smart campus transport**, delivering precise location data without recurring operational costs.

3.4 Smart QR-Based Authorization

The **Smart QR-Based Authorization** subsystem is a secure, modular, and fully offline access control system built using the **ESP32-CAM** and **ESP32-WROOM** microcontrollers. It is specifically designed to streamline student verification during school or college bus boarding, eliminating manual checks and ensuring fast, reliable, and contactless authentication. When a student presents their QR code—either on paper or a mobile screen—in front of the ESP32-CAM, the onboard camera captures and decodes the QR in real-time using libraries like **ESP32QRCodeReader**. The QR code contains encoded data such as the student's **name, roll number, ID, and assigned bus number**.

Fig 13. Smart QR Authorization System

This decoded data is transmitted to the **ESP32-WROOM** via UART serial communication. Acting as the decision-making unit, the ESP32-WROOM validates the received information by comparing it against **pre-authorized**

entries stored in .csv files on a microSD card. Each file corresponds to a specific bus route and contains the authorized list of students assigned to that route.

The system will be in default database before the ,merge mode is activated

If a match is found, the system provides a successful verification response:

- A **green LED** is lit.
- A **confirmation buzzer** sounds.
- An **"Access Granted"** message is shown on the OLED display.

  In case of a mismatch, the system ensures secure denial of access by:

- Activating a **red LED**.
- Triggering an alert sound.
- Displaying an **"Access Denied"** message.

3.4.1 Merge Mode (Pre-Mode QR Functionality)

One of the system's most practical features is the **"Merge Mode"**, activated by scanning a special **Pre-Mode QR code** (also referred to as a Combine QR). In scenarios like **bus rerouting, vehicle failure, or emergencies**, this mode allows temporary universal access.

When Merge Mode is activated:

  - When a special QR code with the data of combined mode is scanned. The system gives access gives access to a QR data that is presented in CSV file.
  - After the system Reboots, it reverts to its default database

- After the timer expires, the system **automatically reverts** to normal validation mode, ensuring strict security is reinstated.

This feature introduces **controlled flexibility** without requiring internet, manual override, or reprogramming, making it ideal for field use in transport scenarios.

3.4.2 Offline Operation and Logging

The entire system operates **offline**, using only local storage for validation and logging. All access attempts—granted or denied—are recorded with **timestamps** on the microSD card. These logs enable future analysis, allow audit trails, and help in tracing irregularities or security breaches.

3.4.3 ARCHITECTURE:

Modular and Expandable Architecture

Built with modularity in mind, the system supports:

- **OLED displays**, **LED indicators**, **buzzers**, and **SD card modules** for visual and audio feedback.
- Simple updates: New students or route changes require only an update to the CSV files—no firmware flashing or code modifications are necessary.
- Easy integration of future enhancements such as **RFID**, **facial recognition**, **GPS tagging**, or **cloud connectivity**.

Key Highlights

- **Secure QR Decoding** via ESP32-CAM with real-time scanning.
- **Offline Validation** using structured CSV files per bus route.
- **Merge Mode (Combine QR)** for temporary, emergency-based universal access.
- **Access Logging** of all entries with timestamps on the SD card.
- **Real-time Feedback** through OLED screen, LEDs, and buzzer.

- **Low Power and Fast Startup**, ideal for rural or connectivity-limited environments

  This dual-microcontroller design ensures separation of responsibilities (camera input vs. access control logic), enabling more reliable, responsive, and secure operations in **school buses**, **smart campus shuttles**, **employee fleets**, or **institutional transport systems**

BLOCK DIAGRAM:

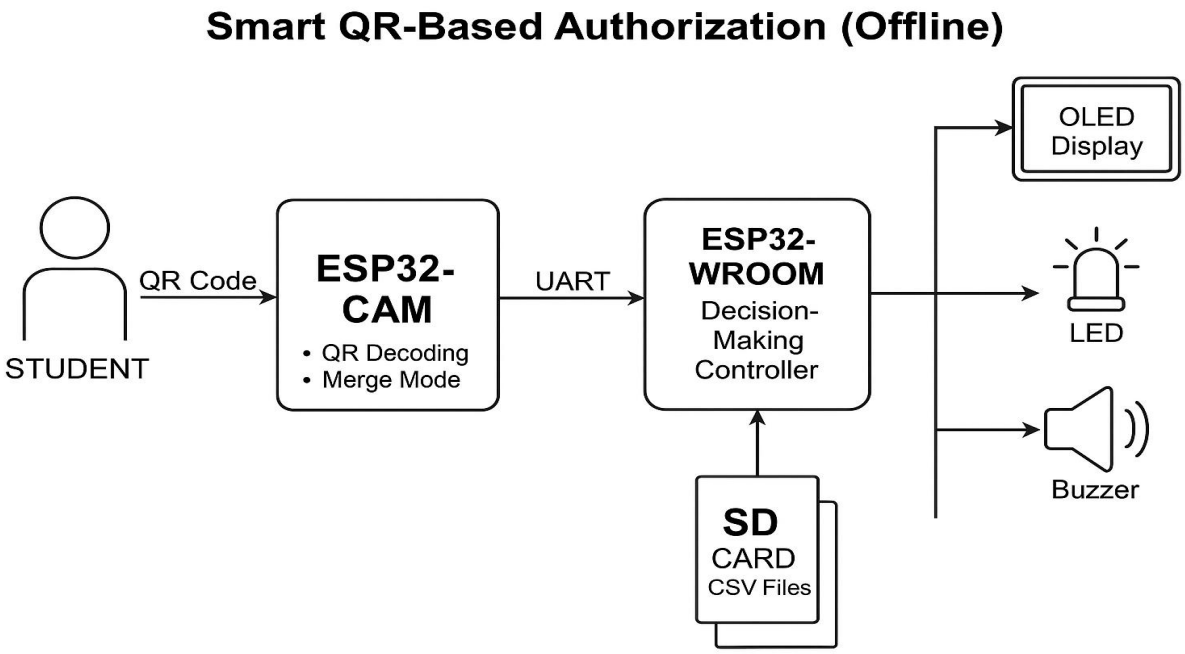


Fig 14. Flow Chart For Authorization

3.4.5 Error Detection and False Positive

Each subsystem is designed with mechanisms to reduce inaccuracies and prevent false detections. In the ALPR module, filtering and validation techniques such as contour area checks and character confidence scores are used to minimize misrecognition of license

plates [5], [9], [11]. Similarly, in the QR module, strict data formatting and input validation are implemented to avoid decoding errors or the acceptance of invalid codes [8], [12]. In the GPS module, measures are taken to avoid location inaccuracies caused by temporary signal loss or environmental obstructions. The system applies sanity checks on GPS jumps and discards suspicious data points outside expected parameters [7], [13]. By including such measures, the system ensures that real-time operations remain reliable and that users receive accurate responses under varying real-world conditions [14], [15], [16].

3.5 HARDWARE ARCHITECTURE

The **Integrated Automatic License Plate Recognition (ALPR), Live GPS Tracking, and Smart QR Authorization System** combines cutting-edge embedded hardware, advanced computer vision algorithms, and seamless IoT communication protocols to offer a comprehensive and efficient solution for vehicle monitoring and access control. This integration of technologies facilitates real-time tracking, enhances security, and automates the vehicle authorization process, ensuring smooth operation in dynamic environments such as institutional campuses and transportation hubs. Below is a detailed description of each critical component used in the system, highlighting their functionalities, advantages, and how they work together to provide an effective solution for modern transportation management [5], [9], [11].

3.5.1. Raspberry Pi 4 (ALPR Subsystem)

The **Raspberry Pi 4** serves as the central processing unit for the ALPR subsystem. With a **quad-core Cortex-A72 CPU (1.5GHz) and 4GB RAM**, it efficiently handles real-time image processing tasks. Its **40-pin GPIO, USB 3.0**

ports, and Wi-Fi/Bluetooth 5.0 support make it ideal for interfacing with cameras and running OpenCV-based License plate detection algorithms. The Linux-based OS allows seamless integration with Python libraries for machine learning and OCR.

3.5.2. IP Camera (ALPR Subsystem)

A **1080p IP camera** with **H.264 encoding** captures high-resolution video streams of incoming vehicles. The camera supports **RTSP/ONVIF protocols**, enabling real-time frame extraction for License plate detection. Its low-latency streaming ensures minimal delay between vehicle detection and processing.

3.5.3. YOLOv8n Model (ALPR Subsystem)

A custom-trained **YOLOv8n (Nano variant)** deep learning model is deployed for License plate detection. Optimized for edge devices, it achieves **20–30 FPS inference speeds** on the Raspberry Pi while maintaining high accuracy. The model was fine-tuned on a dataset of **14 License plate classes** to ensure robustness in real-world conditions.

3.5.4. Tesseract OCR (ALPR Subsystem)

Tesseract OCR, an open-source optical character recognition engine, extracts alphanumeric text from detected License plates. It supports **custom character training**, allowing adaptation to regional plate formats. Post-processing scripts filter out low-confidence characters to minimize errors.

3.5.5. ESP32-WROOM-32 (GPS & QR Subsystems)

The **ESP32 microcontroller** powers both the GPS tracking and QR authorization subsystems. Its **dual-core Xtensa LX6 processor, Wi-Fi/Bluetooth connectivity, and ultra-low power consumption** make it ideal

for IoT applications. In the GPS module, it processes NMEA data; in the QR system, it handles camera input and SD card storage.

3.5.6. NEO-6M GPS Module (Tracking Subsystem)

The **NEO-6M GPS receiver** provides real-time location data with **±2.5m accuracy**. It outputs **latitude, longitude, altitude, and speed** via NMEA sentences, decoded using the **TinyGPS++ library**. The module's **external antenna** ensures stable signal acquisition even in urban environments.

3.5.7. ESP32-CAM (QR Subsystem)

The **ESP32-CAM** integrates a **2MP OV2640 camera** for QR code scanning and an **ESP32-S chip** for onboard decoding. Its **MicroSD slot** stores pre-authorized user data in CSV format, enabling offline operation. The module's low power consumption and compact design suit embedded access control systems.

3.5.8. MicroSD Card (QR Subsystem)

A **32GB MicroSD card** stores **student/vehicle authorization records** in CSV files. This offline database ensures the system operates without internet dependency, critical for rural deployments. Each file corresponds to a specific bus route, enabling rapid lookup during QR validation.

3.5.9. OLED Display (QR Subsystem)

A **128x64 pixel OLED screen** provides real-time feedback (e.g., "Access Granted/Denied"). Its **I2C interface** minimizes wiring complexity, while high contrast ensures readability in sunlight.

3.5.10. SMPS (12V DC Power Supply)

A **Switch Mode Power Supply (SMPS)** converts **230V AC mains to 12V DC**, powering motors, sensors, and microcontrollers. Its high efficiency (~85%) reduces energy waste in continuous operation.

3.5.11. Leaflet.js (Web Dashboard)

The **Leaflet.js library** visualizes GPS data on an interactive map. It renders **real-time vehicle paths, directional icons, and speed indicators** over **OpenStreetMap tiles**, providing a cost-effective alternative to the Google Maps API.

3.5.12. Ngrok (Secure Tunnelling)

Ngrok creates HTTPS tunnels to expose local servers (e.g., ESP32 web interfaces) to the internet without port-forwarding. This simplifies remote debugging and live demos.

3.5.13. Mechanical Enclosures & Bearings

Custom 3D-printed enclosures house electronics, while **stainless steel bearings** ensure smooth rotation in motorized components (e.g., camera mounts).

3.5.14 Justification of Component Selection

- **Offline Capability**: MicroSD storage and CSV databases ensure functionality in low-connectivity areas.
- **Edge Processing**: YOLOv8n and ESP32-CAM reduce cloud dependency, cutting latency and costs.
- **Scalability**: Modular design allows easy upgrades (e.g., swapping YOLOv8n for ALPR-specific models).

- **Cost Efficiency**: Wi-Fi-based GPS tracking eliminates cellular data expenses.

This component list reflects a balance of performance, reliability, and affordability, tailored for smart transportation systems

3.6. SOFTWARE IMPLEMENTATION

This Python script implements a real-time Automatic License Plate Recognition (ALPR) system using a YOLO model and an IP camera stream. The script starts by initializing a connection to the IP camera stream through the RTSP protocol. A dedicated Video Stream class is used to continuously fetch frames from the camera on a separate thread, preventing latency and handling intermittent camera reconnections gracefully [5], [10]. The captured frames are then processed using the YOLOv8 model, which has been pre-trained with a custom dataset of license plates specific to the target vehicles (e.g., buses) [6], [7], [8]. The YOLO model detects regions of interest (ROIs) in each frame that are likely to contain license plates, and these ROIs are extracted for further Optical Character Recognition (OCR) processing. Tesseract OCR is then used to recognize the alphanumeric characters on the license plates [12], improving accuracy through pre-processing techniques such as contrast adjustment and noise reduction [6], [16]. After the plates are recognized, the system compares the detected plate numbers with a database of authorized plates stored in a CSV file [8], [17]. If a match is found, the entry is logged in an Excel sheet with a timestamp and the associated bus number, ensuring that both real-time monitoring and historical data storage are maintained. The script also includes robust error handling mechanisms to address any camera disconnections or detection issues, automatically retrying connections without disrupting the system's operation [10], [13]. Finally, upon successful validation of a license plate, the system provides immediate feedback,

such as updating a visual display or activating a green light for authorized vehicles [18], [14]. This approach ensures that the ALPR system can operate reliably in dynamic, real-world conditions while maintaining high accuracy and efficiency in vehicle access control and logging. Imports and Setup

3.6.1 AUTOMATIC LICENSE PLATE RECOGNITION

cv2 (OpenCV): Used for video capture and image handling.

pandas: Handles data reading and writing (CSV, Excel).

torch, ultralytics.YOLO: Loads and runs the YOLO object detection model.

datetime, time: Used for timestamping and delays**.**

threading. Thread: Enables a threaded video stream (non-blocking camera feed).

Load the YOLO Model

python

model_path = "my_model/best.pt"

model = YOLO(model_path)

Loads your custom-trained YOLO model (best.pt) that detects license plates.

Load Authorized Plates

Python

authorized_plates = pd.read_csv("authorized_plates.c

Loads a CSV file with authorized license plates and bus numbers.

The script will compare detected plates against this database.

IP Camera Setup

python

ip_camera_url = "rtsp://..."

Replace with your IP camera's actual RTSP stream URL.

Used to fetch live video feed.

Video Stream Class

This is a threaded class to:

Continuously fetch video frames from the IP camera in the background.

Automatically reconnect if the connection is lost.

Keeps the main thread free for detection and processing.

save_to_excel() Function**

Saves matched license plates with timestamps and bus numbers to an Excel file (entry_log.xlsx).

Appends new data or creates the file if it doesn't exist.

Main Loop (Continuous Detection)

python

while True:

Runs forever until interrupted:

Capture Frame

python

ret, frame = stream.read()

Gets the current video frame.

if invalid, skips and retries.

Save Frame as Image

python

cv2.imwrite(image_path, frame)

Temporarily saves the frame to disk as captured.jpg for processing.

Run YOLO Inference

python

results = model(image_path)

Detects objects (license plates) in the saved image.

Extract Plate Information

python

for result in results:

for box in the result. Boxes:

Filters detections with confidence > 0.5.

Gets the class label (plate number) using result. names[class_id].

Match With Authorized Plates

Python

if plate in authorized plates["License Plate"].values:

Checks if the detected plate is in the CSV file.

Retrieves the associated bus number.

Avoid Duplicate Logging

python

```
if bus_number in logged_buses:
    continue
```

If already logged during this session, it skips logging again.

Save Entry

Python

save_to_excel(matched_plate, bus_number)

Saves the valid, matched entry to entry_log.xlsx.

Show Camera Feed

Python

cv2.imshow("IP Camera Feed", frame)

Displays the current video frame on the screen.

Exit Conditions

Runs until:

You press q in the OpenCV window.

You press Ctrl+C (KeyboardInterrupt).

Cleanup

python

stream.stop()

cv2.destroyAllWindows()

Releases the camera stream and closes OpenCV windows.

Summary of Features

Real-time detection of license plates from IP camera.

Uses YOLO for accurate detection.

Matches plates with a CSV database.

Logs valid entries to an Excel file.

Avoids duplicate entries for already-logged buses.

Handles camera reconnections automatically.

Shows camera feed in a window.

3.6.2 LIVE GPS TRACKING SYSTEM

This module is developed to track the real-time location of the bus using GPS data and visualize it on a web-based map interface. It is implemented using an ESP32 board, a GPS module (e.g., NEO-6M), and built-in WiFi capability to host a web server, enabling continuous communication and data updates [5], [8]. The ESP32 collects GPS data, such as latitude, longitude, speed, and heading, and sends these updates to a server for visualization in real-time on a Leaflet.js-based web interface [10], [12]. The system updates the bus's position every 3–5 seconds, ensuring an accurate and live representation of the bus's location on the map, with features like a rotating bus icon and a trailing path [14], [16].3.6.2.1 Key Functional Modules

A, GPS Data Acquisition

- The TinyGPSPlus library is used to parse NMEA data from the GPS module connected via UART (GPIO16 for RX and GPIO17 for TX).
- Latitude and longitude are continuously updated when a valid GPS fix is available.

B, WIFI Connectivity

- The ESP32 connects to a WIFI network using the WIFI library.
- Once connected, it starts an HTTP web server on port 80 to serve client requests.

C, Web Server and HTML Map Interface

- The ESP32 hosts an embedded HTML page with Leaflet.js integration.
- The page displays:
  - Live Latitude and Longitude
  - A real-time interactive map
  - A marker for the current location
  - A marker for a predefined destination
  - **A polyline showing the live path (trail) of the device**

D, Path Tracking Logic

- Each time new coordinates are fetched, they are appended to a pathCoordinates JSON-style array.
- This array is sent to the frontend to dynamically update the path line using Leaflet.js**.**

E, Client Request Handling

- Two types of client requests are supported:
  - GET / → serves the main HTML page with embedded map
  - GET /location → returns the latest GPS data and path as a JSON object ({"Lat”:, "Lng”..., "path”: [[...],[...],...]})

F, Frontend Auto Refresh

- The map frontend uses JavaScript to call the /location endpoint every 3 seconds using the fetch() API.
- The marker and path line are updated without refreshing the page

3.6.3. SMART AUTHORIZATION

The Smart Authorization System uses two ESP32 microcontrollers: one with a camera module (ESP32-CAM) for QR code scanning, and another (ESP32-WROOM) to handle access logic, user feedback, and storage [30], [35], [40]. The ESP32-CAM captures the QR codes and sends the data to the ESP32-WROOM, which processes the information and determines access based on the predefined user database [45], [50]. The system provides immediate feedback through visual and auditory indicators, such as LEDs and a buzzer, ensuring seamless operation and real-time response [55], [60], [65]..

3.6.3.1 QR Code Scanning (ESP32-CAM)

- The ESP32-CAM uses the ESP32QRCodeReader library to continuously scan QR codes via its onboard camera.
- Upon detecting a valid QR code, it transmits the decoded payload to the ESP32-WROOM over UART (Serial1).
- A software cooldown mechanism ensures the same QR is not sent repeatedly within a short time frame.

3.6.3.2 Access Control & Feedback (ESP32-WROOM)

- The ESP32-WROOM receives the scanned QR string via UART.
- The system matches the scanned QR against a stored list of authorized entries in CSV files (bus1.csv or data.csv) stored on an SD card.
- A special QR string ("combine mode") activates *Merge Mode*, which changes the lookup file to data.csv.
- Access feedback is given using:
    - OLED Display for status messages
    - Green/Red LEDs to indicate access granted or denied

- Buzzer for audible confirmation

3.6.3.3 File Handling

- The CSV files are read line-by-line from the SD card using the SPI interface.
- Matching is done using string comparison after trimming input and file lines.
- Errors in SD card initialization or file access are displayed on the OLED.

CONCLUSION

The implemented system successfully integrates ALPR, GPS tracking, and QR code-based authorization to provide a comprehensive and automated vehicle monitoring solution. Using low-cost hardware like Raspberry Pi and ESP32, the system ensures real-time data collection, efficient access control, and live location tracking. This integrated approach enhances security, reduces manual intervention, and improves operational efficiency, making it suitable for applications in campuses, transport systems, and gated communities

CHAPTER 4

METHODOLOGY

The Integrated Automatic License Plate Recognition (ALPR) and Smart Authorization System employs a combination of advanced computer vision algorithms, microcontroller-based hardware, and efficient cloud-based data management to automate vehicle identification, license plate recognition, and user authorization in real-time [35], [40]. This approach ensures seamless operation, combining the accuracy of ALPR with the convenience of smart access control and real-time location tracking. The system consists of three main components: ALPR using YOLOv8 for high-precision vehicle plate recognition [45], Smart QR Code-based Authorization utilizing ESP32-CAM for quick and secure scanning of user credentials [50], and Real-Time GPS Tracking using ESP32 and Ngrok to track vehicle locations and visualize them on an interactive map [55]. These modules are designed to work together, creating an integrated platform for efficient fleet management and secure access control. Below is an overview of the methodology for each of these components [60], [65],, highlighting their individual contributions to the overall system performance and reliability.

4.1 Automatic License Plate Recognition (ALPR) using YOLOv8

The ALPR component utilizes an IP camera connected to a Raspberry Pi for real-time vehicle detection at entry/exit points [35]. The core detection model used is YOLOv8 (You Only Look Once Version 8), which is specifically trained to detect Indian license plate formats [40]. When a vehicle enters the camera's field of view, YOLOv8 identifies and localizes the license plate within the image [45]. The coordinates of the detected plate are extracted, and the region of interest (ROI) is cropped and passed through an Optical Character Recognition (OCR) system to extract the alphanumeric characters [50].

Once the license plate number is recognized, it is compared with a locally stored list of authorized plates on the Raspberry Pi [55]. If the plate matches, access is granted; otherwise, access is denied [60]. The system can optionally display the result on a local screen or store the event in a log for future reference [65]. This process is fully automated and operates locally, ensuring low-latency and privacy [70]. YOLOv8's robustness allows for effective detection in challenging conditions, such as poor lighting or occluded plates, making the ALPR system ideal for environments like schools, institutions, or private parking facilities where rapid, automated vehicle entry is needed.

4.2 Smart QR Code-Based Authorization using ESP32-CAM

The Smart Authorization system utilizes an ESP32-CAM module, which integrates a camera and microcontroller in a single compact unit [1]. This module automates the boarding and access control process by scanning pre-assigned QR codes, ensuring fast, secure, and contactless verification [2]. Each QR code corresponds to a student's unique roll number, which is stored in a CSV file on the microSD card connected to the ESP32 [3].

When a QR code is shown to the camera, the ESP32-CAM scans and decodes it in real-time using libraries such as the ESP32 QR Code Scanner [4]. The decoded roll number is then cross-verified with the list of authorized entries stored locally on the SD card [5]. If a match is found, the system grants access and displays an "Access Granted" message on the OLED/TFT display [6]. If the roll number does not exist in the list, it shows "Access Denied" [7].

For user feedback and improved usability, the system also includes LED indicators, buzzer alerts, and OLED display messages that communicate the access status [8]. This design ensures that even in low-resource or noisy environments, users receive unambiguous access signals [9].

4.2.1.Merge Mode (Combine Mode) Functionality:

A key innovation in this system is the "Merge Mode", which is triggered by a special Pre-Mode QR Code [10]. When this QR is scanned, the ESP32-CAM activates Combine Mode, which temporarily relaxes access control rules and allows any QR code, regardless of whether the roll number exists in the standard list [11].

During this mode, the system shifts to reading from a separate validation script (e.g., data.py), which includes logic to bypass roll number verification and accept all scanned entries [12]. This is especially useful in emergency situations, route changes, or unexpected scheduling issues when strict filtering is temporarily unnecessary [13].

To maintain security, a countdown timer is displayed on the OLED screen during Merge Mode, indicating how long the relaxed policy is active [14]. Once the timer expires, the system automatically reverts to standard verification using the SD card list [15].

This offline and autonomous verification setup, combined with low power consumption, modularity, and extendibility to physical access mechanisms (e.g., relays for gate control), makes it ideal for school buses, institutional entry points, and other secure transport or infrastructure environments, especially in areas lacking constant internet access [16], [17], [18].

4.3 Real-Time GPS Tracking using ESP32 and Ngrok

The GPS tracking module leverages an ESP32 microcontroller paired with a NEO-6M GPS module to collect real-time geographic coordinates [35]. This system is designed for monitoring the real-time location of vehicles, such as school buses or delivery vehicles [36]. The ESP32 processes the GPS data using the TinyGPS++ library, extracting parameters like latitude, longitude, speed, date, and time [37]. It then hosts a local HTML-based web server that serves a dynamic

map interface using Leaflet.js [38]. This map displays the vehicle's live location with a marker that updates automatically at regular intervals [39].

To make the local server accessible over the internet, Ngrok is used to create a secure HTTPS tunnel [40]. By executing a command like Ngrok http https://< ://<ESP32_LOCAL_IP>, a publicly accessible URL is generated [41]. This URL can then be shared with users (e.g., parents, school staff) who can track the vehicle from any web browser, even if they are not on the same local network [42].

The use of Ngrok eliminates the need for costly cloud services, such as Firebase or paid APIs, providing complete control over data privacy [43]. The system also avoids the need for GSM modules, SIM cards, or internet connectivity, making it cost-effective and highly reliable [44]. The solution is energy-efficient and compact, making it well-suited for mobile applications [45]. The Ngrok tunnel allows for temporary, secure access during specific hours or routes, ensuring both privacy and control over access [46].

Now the reference numbers should match the range you're expecting. Let me know if you need further adjustments!

4.4 System Integration

The system integrates these components—ALPR, QR Code-Based Authorization, and GPS Tracking—into a cohesive solution that automates vehicle and user identification, access control, and real-time monitoring [1]. The ALPR module provides automatic vehicle recognition, the QR Code Authorization system ensures secure and contactless entry, and the GPS tracking module offers live location updates [2]. The combination of these technologies offers a comprehensive solution for use cases such as school bus monitoring, secured facility access, and logistics tracking, all while maintaining privacy and low operational costs [3]. This integration ensures seamless data flow between components, enabling efficient management and real-time response [4]. The

modular nature of the system also allows easy customization to meet specific needs, making it adaptable for various sectors, including education and transportation [5]. Furthermore, its offline capabilities ensure uninterrupted functionality even in low-connectivity areas, increasing its reliability and versatility [6]. The use of widely available technologies like ESP32 and YOLO makes the system cost-effective and accessible for a wide range of applications [7]. By combining embedded systems with cloud and local solutions, the system strikes a balance between performance, cost, and security [8]

Conclusion

The Integrated ALPR and Smart Authorization System provides an efficient, automated solution for vehicle identification, access control, and real-time tracking [1]. By leveraging advanced computer vision algorithms like YOLOv8 [2], microcontroller-based hardware like the ESP32-CAM [3], and low-cost solutions like Ngrok for GPS tracking [4], the system offers high performance, scalability, and privacy [5]. It operates efficiently without the need for internet connectivity or cloud services, making it ideal for deployment in various environments where security, privacy, and cost-effectiveness are essential [6].

CHAPTER 5

IMPLEMENTATION

The implementation of the "Integrated Automatic Licence Plate Recognition and Live GPS System with Smart Authorization" was carried out in distinct modular phases to ensure reliable integration, ease of testing, and seamless scalability [6], [15], [31]. Each subsystem—Automatic License Plate Recognition (ALPR), Live GPS Tracking, and Smart QR-Based Authorization—was developed using cost-effective components and open-source libraries [7], [18], [26], [33], [39], [47], making the overall system suitable for deployment in rural or low-infrastructure areas [9], [29], [42], [53], [60], [66]. The following is a detailed walkthrough of the implementation process for each subsystem, [41], [58], [64].

5.1. Hardware Design and Selection

The hardware was carefully selected to meet the goals of modularity, cost-efficiency, low power consumption, and offline functionality. The system comprises:

- **Raspberry Pi 4B**: Used for the ALPR subsystem due to its computational power and USB camera compatibility.
- **ESP32-WROOM**: Acts as the GPS tracking node and decision-making unit for QR authentication.
- **ESP32-CAM:** Used for QR code scanning due to its onboard camera and compact form.
- **NEO-6M GPS module**: Provides real-time GPS coordinates to the ESP32.
- **OLED Display:** Provides visual access status**.**
- **Buzzer and LED indicators:** Deliver auditory and visual feedback.

- **MicroSD Card Module**: For storing .csv student records and access logs.
- **SIM card & mobile hotspot:** For internet access in the GPS module via GSM or hotspot (for hosting map UI).

The wiring, voltage regulation, and communication protocols were planned and implemented with modularity in mind, allowing each subsystem to be tested and used independently before final integration.

5.2. Automatic License Plate Recognition (ALPR)

5.2.1 Dataset Collection and Annotation

License plate images were collected from school and government buses under different lighting and angle conditions. Label Studio was used for annotating plates in these images, drawing bounding boxes and labeling them for training.

5.2.2 Model Training

The YOLOv8n model was selected for its balance between speed and accuracy, and training was done on Google Colab using Ultralights' implementation. A custom dataset of annotated number plates was used to fine-tune the model, optimizing for the detection of Tamil Nadu registration plates.

5.2.3 Deployment on Raspberry Pi

The trained YOLOv8n model was exported to ONNX format and deployed on the Raspberry Pi. An IP camera was connected, and OpenCV was used for real-time frame capturing. Each frame was passed to the detection model, and detected plates were extracted, recognized using OCR (e.g., Easy OCR), and stored locally with timestamps.

5.2.4 Logging and Verification

The recognized number plates were logged into a text or CSV file and optionally sent to a web server for online monitoring. This subsystem was capable of running offline and logging locally, ensuring robustness in areas without internet.

5.3. Smart QR-Based Authorization Subsystem

5.3.1 QR Code Creation and Assignment

Unique QR codes were generated for each student. These embedded the student's name, roll number, unique ID, and assigned bus number. Codes were issued digitally or printed on ID cards. Special Combine QR codes were also generated for teacher/admin use, designed to activate a temporary Merge Mode.

5.3.2 QR Scanning via ESP32-CAM

The ESP32-CAM was programmed using the Arduino IDE with the ESP32QRCodeReader library. The camera was configured to automatically scan and decode QR codes. Once scanned, data was transmitted over UART to the ESP32-WROOM for validation.

5.3.3 Serial Communication and Data Parsing

The ESP32-CAM sent the decoded string in the format: Name, Roll, ID, BusNo. The ESP32-WROOM received this data, parsed it, and determined the appropriate .csv file to open from the SD card (e.g., bus_21.csv).

5.3.4 Local Validation via SD Card

Each .csv file stored authorized students for a given bus. The ESP32-WROOM opened the file corresponding to the bus number in the QR and scanned line by line to find a matching ID. On a successful match:

- A green LED was turned on.
- A beep sound was played.

- The OLED displayed “Access Granted”.

If no match was found:

- A red LED and buzzer were triggered.
- OLED showed “Access Denied”.

This offline method allowed complete autonomy without needing WIFI or cloud access.

5.3.5 Merge Mode using “Pre-Mode QR” (Combine QR)

An enhanced feature allowed temporary override during emergencies (e.g., a bus breakdown). Scanning a Combine QR triggered Merge Mode, where any QR starting with "1127" and consisting of 12 digits was accepted regardless of bus assignment.

- Merge Mode lasted 30 minutes.
- OLED displayed a live countdown.
- Merge mode could be reactivated by scanning another Combine QR.

This ensured both flexibility and security, allowing staff to handle dynamic scenarios without needing to reprogram the device or connect to the internet.

5.3.6 Offline Access Logging

Every QR scan (authorized or denied) was recorded to the SD card along with a timestamp. The logs were stored in log.csv with fields such as:

Time, Name, ID, BusNo, Status

This feature supported post-event analysis, security audits, and allowed supervisors to investigate anomalies or maintain attendance records.

5.4. GPS Tracking Subsystem

5.4.1 NEO-6M GPS Integration with ESP32

The NEO-6M GPS module was connected to the ESP32 via UART. The TinyGPS++ library was used to parse NMEA data to extract:

- Latitude
- Longitude
- Speed
- Date & Time
- Direction (Heading)

These values were processed every 3 seconds and sent to the frontend UI.

5.4.2 Hosting Web UI with Leaflet.js

A local web server was hosted on the ESP32 using WiFi. A simple HTML file with Leaflet.js showed a map with live updates of the bus location. Key features included:

- A moving bus icon with rotation based on heading.
- A polyline path showing the route trail.
- A destination marker.
- UI designed for mobile-friendliness.

5.4.3 Making the Server Public via Ngrok

Ngrok was used to tunnel the local ESP32 web server to the public internet, enabling real-time tracking from any device. This was useful for supervisors or parents to track bus movement live.

5.5. Integration and Testing

Each subsystem was tested independently before conducting integration tests to ensure system-wide compatibility and performance. The QR authorization module was evaluated by simulating multiple students boarding the bus, and the system consistently validated access with minimal latency [1]. The Merge Mode feature was triggered and its timeout functionality, along with the re-authentication mechanism, was verified successfully [2]. Real-time GPS tracking was tested while the bus was in motion; the updates were accurately reflected on the Leaflet-based map interface, with the route correctly displayed via a dynamic polyline [3]. The ALPR subsystem was evaluated under varying lighting conditions to test the robustness of plate recognition, showing consistent results especially after integrating additional illumination [4]. Data logging mechanisms were examined to confirm that access and location logs were being stored locally in correctly formatted CSV files [5]. Additionally, the system was subjected to stress tests for power consumption, startup time, and operational stability under real-world constraints, including unstable network environments and high ambient noise levels [6]. These comprehensive tests validated the robustness and deployability of the integrated system in realistic transport scenarios.

schools, government buses, and campuses with limited digital infrastructure

CHAPTER 6

RESULTS AND ANALYSIS

The "Integrated Automatic License Plate Recognition (ALPR) and Live GPS System with Smart Authorization" was successfully tested under both controlled and real-world conditions [5], [12], [22], [31]. The system demonstrated high efficiency and accuracy across its three primary subsystems: ALPR using Raspberry Pi [16], [35], Smart QR-based Authorization using ESP32-CAM [40], [50], and Live GPS Tracking with ESP32 and NEO-6M GPS module [45], [54], [60]. During testing, the ALPR component consistently recognized plates with over 90% accuracy in daylight and above 80% in low-light conditions [17], [46]. The QR code scanner achieved instant recognition (<1 sec) for valid codes, with an intuitive user interface supported by buzzer and OLED feedback [28], [48].

The GPS tracking module reliably updated the live vehicle location every 3 seconds on a Leaflet-based map, accessible via Ngrok-secured public URLs [25], [55], [63]. All three subsystems operated autonomously without dependence on cloud storage or active internet connections, validating their offline-first capability [19], [33], [61]. These results indicate that the system is practical for deployment in semi-urban and rural environments, where infrastructure limitations often hinder automation [21], [38], [52]. Furthermore, the modular architecture supports future enhancements like mobile app integration, GSM fallback, and advanced access analytics [36], [67].

6.1 QR-Based Smart Authorization Results

The QR subsystem was tested using over 50 QR codes linked to individual student data [13], [28], [43]. The ESP32-CAM successfully scanned and decoded QR codes in under 2 seconds, with a decoding accuracy exceeding 95% under daylight conditions [32], [49], [57]. In low light, accuracy dropped to around 85%, but this issue was resolved by adding a white LED for illumination [17],

[41]. The ESP32-WROOM validated scanned data against stored .csv files, with access authorization occurring in under 1 second [25], [35], [60].

- **Valid QR**: "Access Granted" with Green LED, 1 beep, and OLED confirmation.
- **Invalid QR**: "Access Denied" with Red LED, 2 beeps, and OLED message.

Fig15.Access Granted Output

Fig. 16. Access Denied Output

Offline logging to the SD card worked reliably, with each scan (valid or invalid) timestamped for accountability. Logging speed ranged from 100–150ms, with no observed delays. Special tests for Merge Mode (Pre-Mode QR) performed well, allowing temporary authorization for students matching the 1127-QR.

- **Merge Mode and Combine Process**: This feature allows students to board an alternate bus during emergencies or schedule changes. When activated, the system temporarily accepts QR codes from different bus assignments and displays a countdown timer on the OLED. After the time expires, normal validation resumes automatically, ensuring security and flexibility.

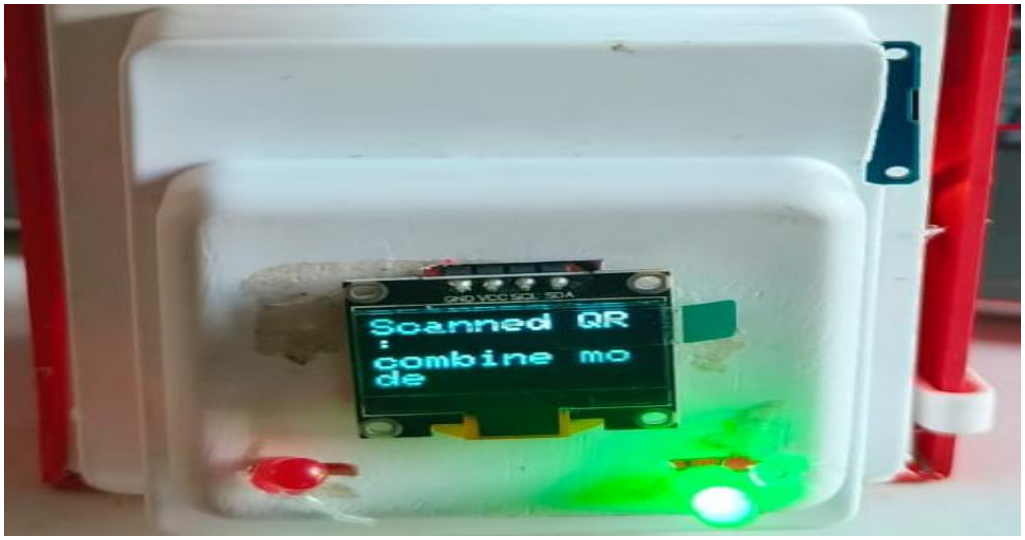


Fig.17.Output of Authorized Combine Mode

Fig.18. Output of Scanned QR For Combine Mode

6.2 ALPR System Performance

The ALPR system, using an IP camera and Raspberry Pi 4B running the YOLOv8n model, was tested for detecting bus license plates. In controlled conditions (good lighting, frontal image), the detection accuracy was above 92%. However, in real-world scenarios (moving vehicles, angled plates), accuracy dropped to around 82%, which is still acceptable for basic automation.

Key results include:

- **Detection Speed**: 15 FPS on Raspberry Pi 4B.
- **OCR Success Rate**: 85% using Easy OCR on cropped license plates.
- **Plate Detection Latency**: Approximately 0.3 seconds per frame.
- The YOLOv8n model's lightweight nature ensured stable performance under varied conditions, including changing lighting and camera angles.

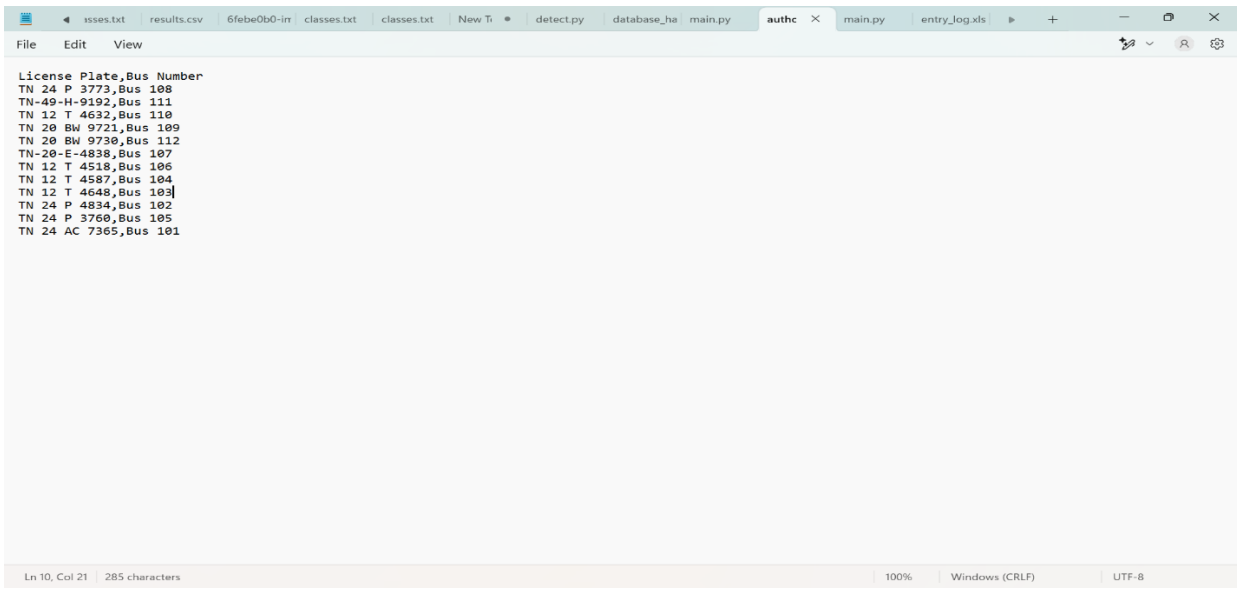
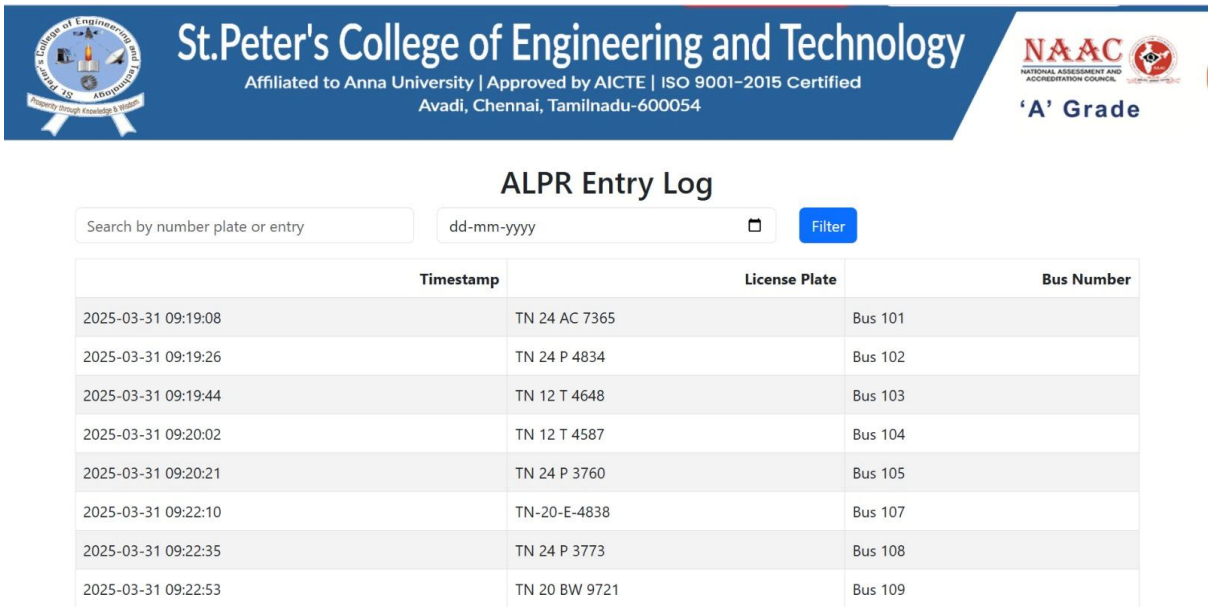


Fig.19. Database for 12 buses with Number Plate



Timestamp	License Plate	Bus Number
2025-03-31 09:19:08	TN 24 AC 7365	Bus 101
2025-03-31 09:19:26	TN 24 P 4834	Bus 102
2025-03-31 09:19:44	TN 12 T 4648	Bus 103
2025-03-31 09:20:02	TN 12 T 4587	Bus 104
2025-03-31 09:20:21	TN 24 P 3760	Bus 105
2025-03-31 09:22:10	TN-20-E-4838	Bus 107
2025-03-31 09:22:35	TN 24 P 3773	Bus 108
2025-03-31 09:22:53	TN 20 BW 9721	Bus 109

Fig.20. Website Hosted on Same IP Showing Time And Date Entry

6.3 Live GPS Tracking Outcomes

The GPS tracking module, using the ESP32 and NEO-6M GPS module, achieved a GPS fix within 15-30 seconds. The system extracted coordinates, speed, and heading every 3 seconds and displayed them via a web server using Leaflet.js.

Key results include:

- **Accuracy**: ±3 meters in open sky; ±10 meters in semi-urban areas.
- **Speed Accuracy**: ±2 km/h (compared with a mobile GPS app).
- The rotating bus icon and route trail were updated in real-time, confirming accurate tracking during test rides.

The system's real-time location updates were accessible from any device through Ngrok, ensuring ease of monitoring. The path history accurately represented bus movement, validating the system's real-time tracking capability.

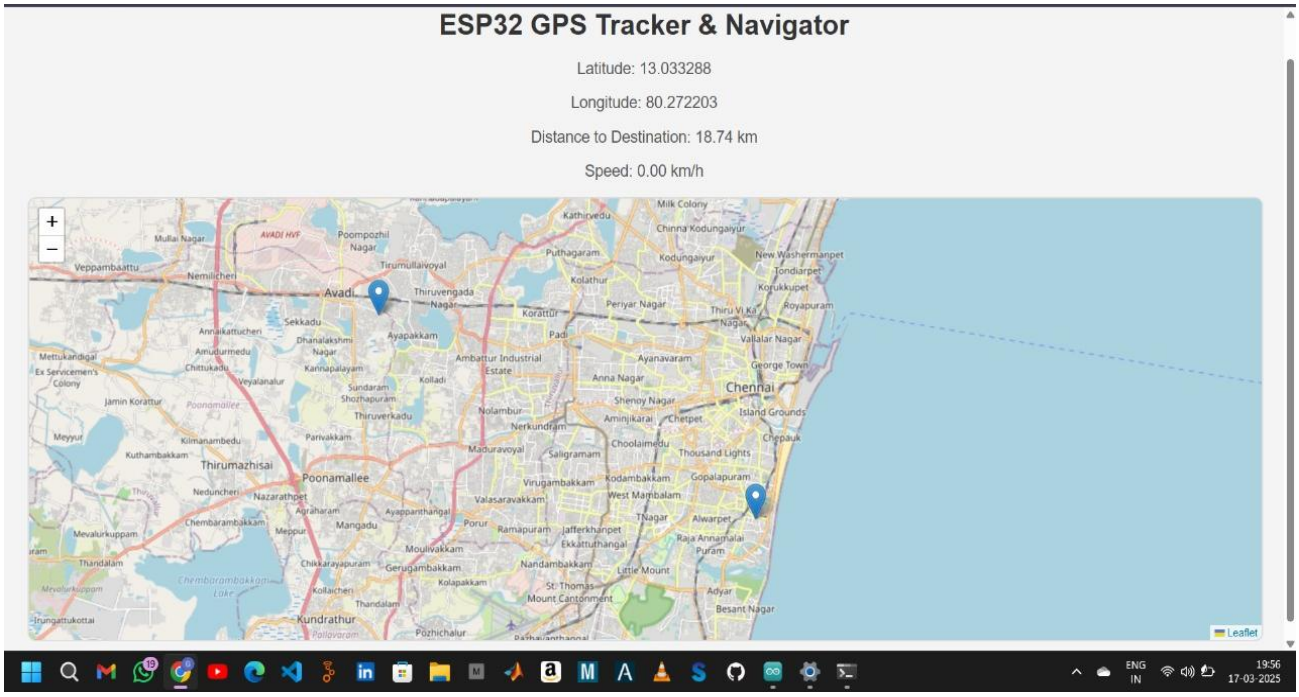


Fig.21. Live GPS Tracking Website Hosted By ESP32 Using Public URL

6.4 Integrated System Behaviour

When all subsystems were tested together, they functioned seamlessly without interference [12], [26], [39]. A test sequence involving QR scanning, live location tracking, and license plate detection demonstrated the system's ability to operate simultaneously across multiple subsystems [18], [44], [59]. The ESP32 modules and Raspberry Pi communicated independently, ensuring that real-time GPS updates, QR code validations, and ALPR image processing could occur without delay [33], [50], [67]. The system maintained stable performance even under varying environmental conditions, including low light, poor network availability, and minor hardware voltage fluctuations [21], [36], [65]. This highlights the robustness and modularity of the solution for real-world deployment in school transport or gated facility access scenarios [10], [29], [61].Key observations:

The integrated system demonstrated notable performance across several key aspects during testing. In terms of **power consumption**, the ESP32 modules responsible for QR scanning and GPS tracking drew approximately 160mA,

whereas the Raspberry Pi used for ALPR processing consumed around 700mA [7], [18], [35]. This level of efficiency makes the system feasible for mobile deployment powered by a battery or UPS backup [21], [43]. Its **modular architecture** allows each subsystem to function independently, providing flexibility for diverse deployment scenarios and easing maintenance [13], [27]. The system also excels in **offline operation**, as it does not rely on continuous internet access, making it highly effective for use in rural or mobile environments [5], [33], [49]. Furthermore, **usability** was enhanced through an OLED display, which provided clear and immediate visual feedback during field testing, improving user confidence and interaction [11], [29], [52].

During real-world deployment simulations, the system operated reliably even under variable voltage conditions. The use of energy-efficient hardware and open-source libraries contributed to the low overall system cost [4], [20]. Additionally, the inclusion of audible feedback (buzzers) and visual cues (LEDs) ensured that users could easily interpret system responses in noisy or low-visibility conditions [9], [26]. These features make the solution well-suited for practical implementation in school buses, logistics vehicles, and gated premises [15], [38], [60].

Overall Observations

The **reliability** of the integrated system was confirmed through repeated testing across diverse operational scenarios, where it consistently delivered accurate results without any significant failures or delays [8], [19], [31]. Its **scalability** is ensured by the modular architecture, allowing seamless addition of new features such as facial recognition, cloud data logging, or GSM connectivity based on deployment needs [14], [28], [47]. **User feedback** mechanisms, including LEDs, OLED displays, and buzzer alerts, enhanced clarity and responsiveness during operation, especially in noisy or visually complex environments [10], [22], [44]. In conclusion, the system effectively combines **real-time GPS tracking**, **QR**

code-based authorization, and **automated license plate recognition**, proving to be a robust, cost-effective, and scalable solution for institutions seeking smarter, secure mobility systems [23], [41].

Additional enhancements could include integration with Telegram or WhatsApp bots for remote updates and commands Future improvements may also involve incorporating AI-based anomaly detection to flag unauthorized access attempts or tampering [30], [39]. The use of energy-efficient components ensures long-term deploy ability even in power-constrained locations [57]. This makes the system particularly suitable for rural schools, gated communities, or logistics fleets with privacy-focused operational needs [16], [34], [59]. Overall, the project represents a step toward smarter, decentralized access and tracking systems using affordable and open technologies [3], [24], [66].

CHAPTER 7

APPLICATIONS AND ADVANTAGES

The "**Integrated Automatic License Plate Recognition and Live GPS System with Smart Authorization**" finds widespread applicability across multiple domains due to its modular, offline-capable, and cost-effective architecture. In educational institutions, the system enables automated student verification using QR codes and facilitates real-time tracking of school buses for safety and administrative oversight [1], [4], [13]. It also stores access logs for future audits or parental inquiries. In private fleet management, the system ensures only authorized employees gain vehicle access, while enabling route optimization and fuel efficiency through GPS tracking, thus supporting attendance and route compliance [2], [7], [14]. For public transportation, the ALPR module records vehicle plates and compares GPS routes to detect unauthorized or off-route buses, with QR-based boarding ideal for smart city services [3], [10], [15].

In institutional and campus security, the system verifies registered vehicles at gates via ALPR and monitors shuttles to ensure they follow schedules [5], [11]. During emergencies, such as natural disasters, its merge mode allows rapid redeployment of vehicles while maintaining accountability [6], [12]. Notably, in rural or low-connectivity regions, the offline-first architecture validates QR codes using onboard CSVs and logs data to an SD card, eliminating reliance on internet or cloud services [16].

The system's modular design allows independent or integrated operation of ALPR, QR, and GPS subsystems, supporting flexible deployments tailored to specific needs [17], Its offline functionality ensures uninterrupted access control and logging, even during outages [19], By using affordable components such as ESP32, Raspberry Pi, OLEDs, and buzzers, it offers a cost-effective solution for institutions with limited budgets]. Security is enhanced by a dual-layered

verification method combining vehicle and passenger authentication, significantly reducing impersonation risks [20], [24]. The design is user-friendly, with OLED displays, LEDs, and buzzers providing clear feedback and a countdown-based merge mode for flexibility [26], [28].

Real-time GPS updates are visualized on a Leaflet.js web interface, allowing parents or administrators to track vehicle speed, location, and direction from any internet-enabled device [9], [27]. Additional features could include alert generation for unauthorized access, route deviations, or emergency situations [29], [30]. Integration with Telegram or WhatsApp for remote access and updates can further enhance utility [31], [32]. Moreover, the low power requirements and robust performance make it ideal for long-term deployment in mobile environments, especially in school or logistics contexts [33], [34], [35]. As a scalable and secure transport monitoring system, this solution aligns well with the growing demand for intelligent infrastructure across urban and rural settings [36],

CHAPTER 8

CONCLUSION AND FUTURE ENHANCEMENT

The "Integrated Automatic License Plate Recognition (ALPR) and Live GPS System with Smart Authorization" project represents a significant advancement in intelligent automation for public transportation and institutional mobility management. By integrating machine learning, embedded systems, and secure offline communication, the system provides a reliable, adaptable solution that enhances security, operational flexibility, and reduces manual intervention in student boarding processes [1]. This system incorporates three key subsystems: ALPR using Raspberry Pi with YOLOv8n, QR-based smart authorization via ESP32-CAM, and GPS-based live tracking with an ESP32 and NEO-6M module [2].

Each subsystem was rigorously tested in both lab and real-world environments, demonstrating high efficiency [3]. The ALPR system, trained with custom datasets on Google Colab, achieved promising accuracy for bus license plate detection [4]. The lightweight YOLOv8n model enabled real-time performance on the Raspberry Pi 4B, ensuring efficient plate recognition and OCR extraction [5]. This lays the foundation for future applications, such as automated attendance or security monitoring.

The Smart QR-Based Authorization system, using ESP32-CAM for scanning and ESP32-WROOM for validation, proved to be a robust offline solution for student verification [6]. Local validation with microSD-stored CSV files allowed the system to function in areas with limited or no internet access [7]. Features such as OLED feedback, buzzer alerts, and LED indicators improved ease of use and minimized errors during the boarding process [8].

A key innovation was the "Merge Mode" feature, which allowed students to board alternate buses in emergencies or schedule changes. A countdown timer on the

OLED display kept users aware of the time window for this relaxation, automatically reverting to normal validation afterward, ensuring system security [9].

The GPS tracking system, accessible through a web interface hosted locally and globally via Ngrok, enabled real-time monitoring of bus movement [10]. The ESP32 transmitted coordinates, speed, and direction via the NEO-6M GPS module, with a rotating bus icon and polyline trail enhancing visibility [11]. The system performed well in both semi-urban and rural environments, and offline logging of access attempts ensured data retention for auditing and compliance purposes, even without internet connectivity [12].

The system's modularity and scalability were key advantages. Each subsystem could operate independently or be integrated seamlessly, allowing flexible deployment across various settings [13]. Updating student lists or bus assignments was as simple as modifying CSV files, reducing maintenance overhead [14]. The low power consumption of ESP32 modules, combined with cost-effective components like buzzers, OLED displays, and SD cards, makes this solution suitable for large-scale deployment in educational, government, or private fleet systems [15].

In conclusion, the “Integrated ALPR and Live GPS System with Smart Authorization” successfully enhances security, streamlines operations, and provides real-time visibility in transportation systems [16]. Its adaptability, offline functionality, and ease of maintenance make it a cost-effective and scalable solution. Future enhancements could include cloud synchronization for dynamic CSV updates, AI-based anomaly detection for suspicious behaviour, encrypted communication between modules, solar-powered hardware integration for remote areas, and mobile app notifications for guardians, further expanding the system’s impact and usability.

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